UNIVERSITY OF CALIFORNIA
SANTA CRUZ

ECE 129 Capstone Series

WILDLIFE AUDIO PLAYBACK

Senior Capstone Report Submitted in Partial Satisfaction of the Requirements for the Degree of

Bachelor of Science

in

ELECTRICAL ENGINEERING

by

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Abstract

This senior capstone project aims to create a device that will be used to study the effects of human disturbances on large fields in Central Africa using grid-style audio playback experiments. This project was proposed to the team by UCSC Environmental Studies graduate student Mike Kowalski, who was searching for students to redesign audio playback devices he created and attempted to use in the summer of 2023. The redesigned audio playback device addresses issues Mike faced last summer by incorporating features including extended battery life, field-adjustable settings, solar charging, and robust environmental protection. The new device created by this team will allow Mike and other environmental scientists to obtain new data on wildlife's reaction to human disturbances and potentially influence policies for managing multi-use landscapes and conservation areas.

1 Introduction [Aiden / Liam]

1.1 Motivation

As a single species, humans have the greatest impact on the planet - we have altered over 3/4 of the Earth's surface, changed the global climate, and modified our local environments to suit our needs. Most would consider human disturbance to be a primarily negative force on wildlife across the globe. In reality, there are various forms of human disturbances. Repulsive and lethal disturbances include (but are not limited to) deforestation, pollution, urbanization, and overexploitation of wildlife. Activities such as hunting regulation, non-invasive research, and noise management can prove to have a neutral impact on wildlife. However, disturbances such as habitat restoration, supplemental feeding, sustainable agriculture practices, and conservation and protected areas can be attractive and beneficial to the wildlife involved.

In particular, protected and wildlife conservation areas have been often viewed as having numerous positive impacts on wildlife. These types of areas preserve natural habitats, support endangered species, conserve the natural ecosystem, and promote education and awareness of positive disturbance practices. These types of areas are commonplace in Central Africa countries, where 25% of their northern savannahs are protected areas, but are being increasingly encroached upon by settlements and livestock grazing [1]. The growing population of Africa and the invasion of these landscapes causes them to convert to multi-use landscapes. Multi-use landscapes do not follow the same policy as conservation areas, and allow for the degradation of habitat, cause stress and harm to the wildlife within them, and increase in extinction risk of particular species.

As more and more of these areas change, there becomes an increasing demand to study the effects of human behavior on the wildlife being invaded. Our project, in particular, is to observe the impacts of human disturbance on large felids in Eastern Africa, as well as the broader ecosystems involved. Observing the reaction wildlife has to our human disturbance can reshape the policies of multi-use landscapes and how they are managed moving forward. To do this, grid-style audio playback experiments are the most practical way to simulate human presence without causing harm to the wildlife being studied.

1.2 Background

The concept of grid-style audio playback experiments to observe wildlife behavior was presented to the team in a Fall quarter presentation in ECE 129A by UCSC environmental studies graduate student Mike Kowalski. He drew inspiration from the Santa Cruz Puma Project, which conducted similar experiments on mountain lions in the Santa Cruz Mountains. These types of wildlife observation experiments involve using this speaker grid system in conjunction with telemetry collars and game cameras to observe large felids' behaviors, as shown in **Figure 1.2.a**.

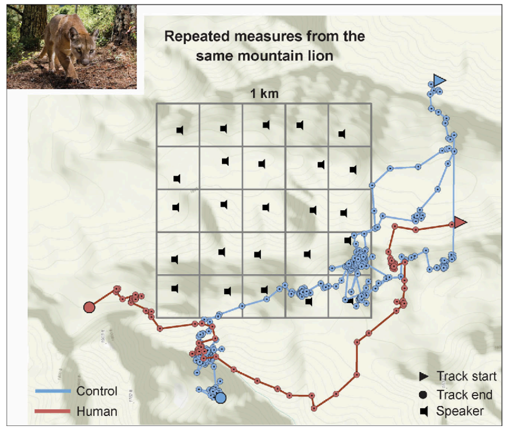

Figure 1.2.a: Previous experiment playback grid by the Santa Cruz Puma Project observing mountain lion reactions to human and frog noises. Suraci et al., 2019.

Mike had previously conducted a similar experiment to the one shown in **Figure 1.2.a** Uganda during the summer of 2023 to observe the behaviors of lions and leopards. He set up a 16 square kilometer grid in the savannah consisting of 25 speakers and 12 game cameras, attempting to gather data on the felids' reaction to human disturbance. The speakers used in his experiment played three different sound treatments: human, cattle, and a control (frogs), each at 80 dB SPL at one meter. The three treatments contained ten tracks, with five randomized orders of those tracks spread amongst the 25 speaker systems. The treatments played at a forty percent duty cycle, meaning a five-minute track would play followed by seven and a half minutes of silence. The devices Mike used to perform this experiment consisted of a six volts and five amp-hours sealed-lead-acid battery, an MP3 player, a DC-DC step-down converter, a waterproof speaker, and a waterproof Pelican Case as shown in **Figure 1.2.b** below:

Figure 1.2.b: Mike's playback speaker design used in his 2023 experiment in Uganda.

When Mike attempted this experiment, he ran into several problems getting his device to function properly. The most prevalent issue encountered was a short battery life of around ten hours, making it nearly impossible for him to replace all the batteries frequently enough to keep the grid functioning. This also required Mike to disturb the wildlife when changing the batteries, obscuring the data gathered. Batteries also overheated during the experiment due to the strong African sun, and animals damaged the devices by chewing exposed wires, rendering him unable to conduct his experiment and gather the necessary data for his research. Another less impactful, but still important issue Mike ran into was that silence was hard coded into the audio treatments, meaning there was only a single track as a treatment, so he could not adjust the duty cycle of the speaker output. He also could only store one audio file on each cheap MP3 player, which made it inconvenient to change the treatment because he would have to go out into the field and change the MP3 itself. An unsuccessful experiment caused by these numerous issues called for a complete redesign of the audio playback device.

2 Wildlife Audio Playback Device [Aiden / Liam / Kamran/Cole]

2.1 Project Deliverables

The solution to Mike's failed experiment was to redesign and build upon the previous design to account for all issues encountered during his first trial, as well as anticipate more that may arise. To achieve this, Mike presented us with several deliverables. These included:

1. A minimum of seven days of battery life to prevent the need to interrupt the experiment and change the battery in the field.
2. The ability to conveniently change the audio volume and treatment parameter in the field without significant disassembly or tools required.
3. 80 dB Sound Pressure Level (SPL) audio output at one meter from the device.
4. The capability to withstand the harsh environment of Kenya, including dust, rain, wildlife, and unwanted tampering of the device.
5. A lightweight and compact design for portability and the ability to be conveniently mounted in trees.
6. The ability to upload a minimum of three treatments containing ten audio tracks, with a total of thirty tracks.

To achieve these, the team came up with several solutions in collaboration with Mike. Our first was a higher quality battery with a higher energy density and resilience to heat encountered while inside the box for extended periods. Solar panel charging capabilities were also chosen to allow the battery to recharge during the day, extending the life of the battery indefinitely. To address volume and treatment adjustment, we implemented a microcontroller with three buttons connected to allow the user to raise and lower the volume and scroll through treatments available. The microcontroller would also have sleep modes programmed in to sleep in between tracks being placed, vastly reducing the amount of power used during its duty cycle. An MP3 Player with a maximum of 32 GB of storage was connected to this microcontroller to provide ample space for treatments and tracks in the device. Finally, a lightweight, robust casing, following NEMA Enclosure Type 4 Standard was chosen and painted in a light, natural color to withstand the environment of Kenya and blend into the surrounding foliage [2].

2.2 Proposed Design

Our proposed design followed the block diagram shown in **Figure 2.2.a** below:

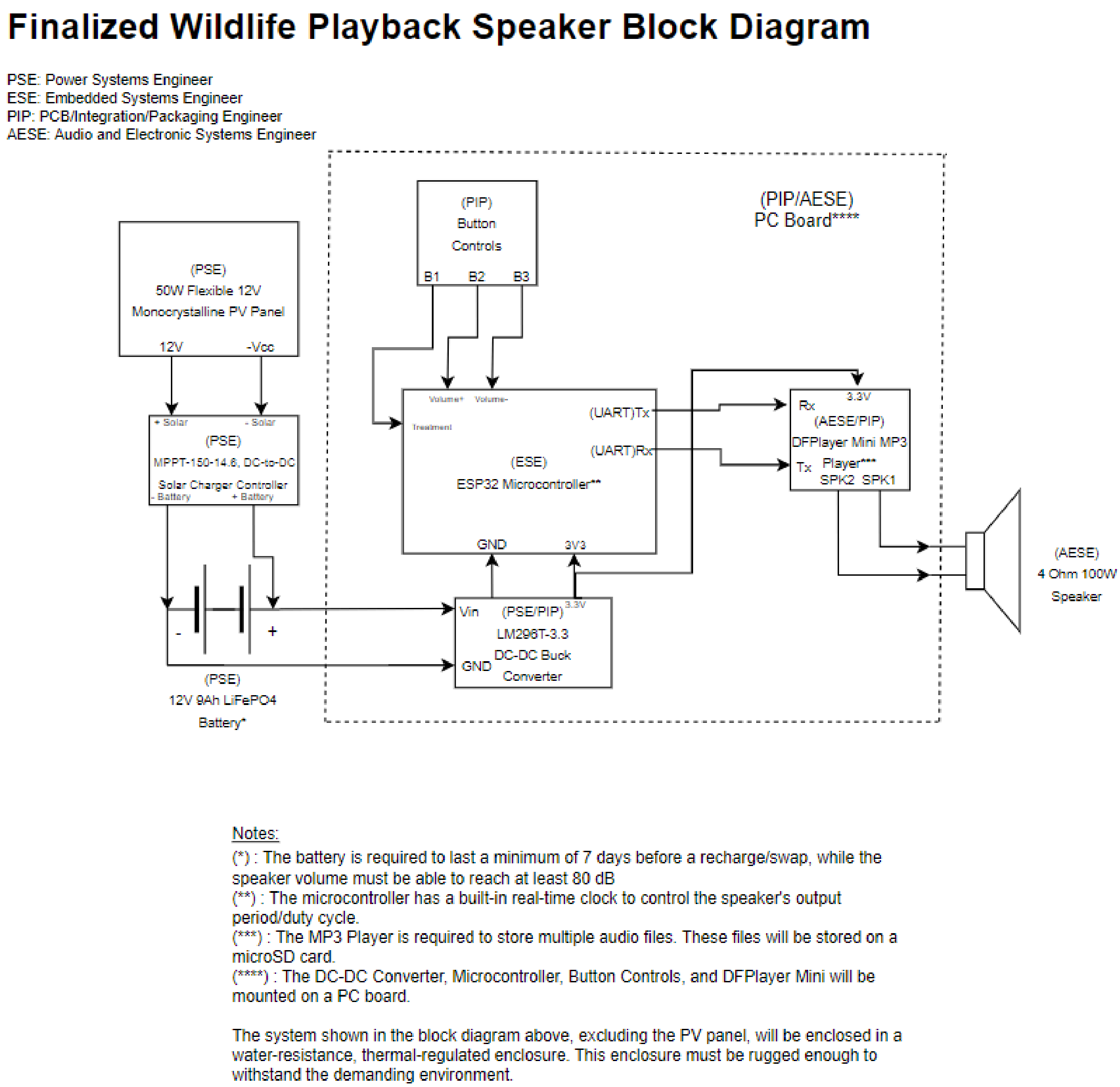

Figure 2.2.a: System Level Finalized Block Diagram of Audio Playback Device.

The design shown above consists of four main subsystems: Power System, Embedded System, PCB and Packaging System, and Audio and Electronics System. The Power System is primarily focused on delivering a device battery longevity of more than seven days. To do this, a 50W 12V monocrystalline photovoltaic panel is used to charge a 12V 9Ah LiFePO4 battery, which is controlled by a Maximum Power Point Tracing 150W 14.6V DC-DC solar charge controller. The battery is connected to an LM2596T-3.3 DC-DC buck converter switching regulator to step down the nominal voltage of 12V from the battery to 3.3V to power the rest of the circuit. The embedded systems are focused on delivering user adjustability of the device and controlling the audio output periods. To accomplish these requirements, the device is controlled by an ESP32-S3-WROOM-1 microcontroller which uses its ADC pins in conjunction with its serial communication pins to control the treatment being played and the volume of the device. Physical buttons are used with these microcontroller functions to allow user adjustment of the device. The audio and electronics system is responsible for delivering an audio output of 80 dB SPL with client-approved audio quality and sufficient audio storage for 3 treatments containing 10 tracks each. To fulfill these requirements, the audio files are stored on a 32GB micro SD card on the DFPlayer Mini MP3, which supports up to 100 folders, each supporting up to 255 audio tracks. The microcontroller uses its UART communication pins to control when the MP3 player is playing, adjust the volume, and change audio treatments/folders. The MP3 Player contains a 3W audio amplifier that drives a 4Ω 50W RMS waterproof high-sensitivity speaker. The PCB and Packaging system is responsible for providing electronic component integration on a printed circuit board and ensuring that the device can withstand the local climate where it will be used. The button controls, microcontroller, buck converter, and MP3 player are mounted on a 2.5” x 4” printed circuit board. The PCB, battery, and charge control are contained in a 10.6” x 6.1” x 3.9” NEMA Type 4 Enclosure protective case with the speaker mounted on the top side of the case.

2.3 Validation of Requirements

To implement our design requirements, we provided several options of solutions for each, which was then narrowed down based on testing and review by our client Mike. The final design decisions that met our design requirements are as follows:

For deliverable 1, multiple tests were conducted to ensure that our device would last the intended amount of time. The first validating test conducted was a full system test. For this test, we connected the battery to our finalized device. We then played the human noises track at a duty cycle of 60%. The reason the human noise track was played was because it consumes the most power. This track was run at a duty cycle of 60%. From this test, our device was able to last 6 days, which is one day under our 7-day goal. However, this test was run at a 60% duty cycle instead of Mike’s desired 40%, and did not include the solar panel. Solar panel testing was also conducted and showed that with a solar panel, the battery could be fully charged in 2 hours when there was a UV rating of 7. Under full load, the solar panel was able to charge a battery on days with a UV rating of 4 or higher. In Kenya, the average UV rating during July, August, and September (deployment dates) is 5. This means that as long as the device has one day with a UV greater than 4, it will recharge. With a solar panel, the device can last significantly longer than 7 days, meeting our deliverable. For more information on this deliverable, refer to sections 3.1.4 and 3.1.5.

For deliverable two, requiring the ability to change volume and track selection without significant product or code disassembly was satisfied by the integration of external buttons into the device in the embedded system. The microcontroller converts digital inputs from the external buttons and translates them into commands from the board to the MP3. Users can then use button 1 to change tracks, use button 2 to increase the volume, and use button 3 to decrease the volume.

As for deliverable three, requiring an 80 dB SPL audio output level, we used an SPL Meter to record each of our treatments maximum and average dB levels. We verified that each treatment reached a maximum of 80 dB and above. The average dB level was recorded for interest purposes, despite the deliverable only asking for the ability to reach 80 dB. Shown in **Table 2.3.a** below are the measured average and maximum audio levels of the three treatments:

Measurement	Control (birds)	Human Voice (Swahili)	Cattle
Average (dB)	69.1	76.6	64.9
Maximum (dB)	87.7	83.5	81.2

Table 2.3.a: Measured Average and Maximum Audio Levels of Control, Human Voice, and Cattle Treatments

For deliverables four and five, requiring a robust, lightweight case that is portable and can withstand the harsh environment of Kenya, we found a case that follows the NEMA Type 4 Enclosure Standard. The Standard for NEMA Type 4 Enclosures states that "Enclosures constructed for either indoor or outdoor use to provide a degree of protection to personnel against access to hazardous parts; to provide a degree of protection of the equipment inside the enclosure against ingress of solid foreign objects (falling dirt and windblown dust); to provide a degree of protection concerning harmful effects on the equipment due to the ingress of water (rain, sleet, snow, splashing water, and hose directed water); and that will be undamaged by the external formation of ice on the enclosure. [2]" After extensive research, we found a case that was verified to comply with this standard. The most important aspect of the standard was the degree of protection from the ingress of water, so we opted to test this by spraying our device with a shower hose. We found after twenty seconds of direct spraying on all sides that no water was allowed into the device, meeting our requirement. We chose a size based on the dimensions of each of our components, to ensure each fit with ample room to not come in harmful contact with one another, while still keeping the size as low as possible. The only issue we ran into was color, as we could not find a case that came in a natural color such as green or tan that was the right dimensions to mount all of our components to. Our proposed solution suggested by the client regarding this issue was to repaint it a tan color. Holes were drilled into the sides to allow for the mounting of a python lock, which would be used to mount the case to trees as well as prevent unwanted tampering.

For deliverable six, requiring the project to have the ability to have at least three treatments containing ten audio tracks, the DFPlayer Mini MP3 Player was selected to store and play the audio tracks. The DFPlayer Mini can store up to 100 folders/treatments, each folder with each folder able to hold up to 255 audio files. Treatments can be uploaded onto a micro SD card and then inserted into the slot on the DFPlayer Mini, making it a simple and modular design with ease of changing audio treatments and tracks if needed in the field.

3 Subsystems Design and Verification

3.1 Power Systems [Kamran Pakravan]

3.1.1 Power Systems Introduction

The power systems engineer is a crucial role that must be adequately fulfilled to ensure the success of this project. The role of the power systems engineer for this device was given to Kamran Pakravan. The power systems engineer is in charge of designing a power system that will effectively meet the power needs of the project. In the case of this project, the client-specified requirement of the power engineer is to ensure that the final device can run a minimum of 7 days at a duty cycle of 40%. The power system must also be resilient enough to withstand the conditions of the deployment environment in Kenya.

Tackling this problem took place through analyzing the underlying issues of the previous prototype produced by the client, Mike, and addressing them. Once the issues had been found, research on the most optimal power components was conducted, and engineering standards were analyzed to ensure that the components selected met performance and quality standards.. Once the research was completed, the power engineer moved on to the component selection process. A power budget was then created to effectively map out the power consumption of each component inside the device. The power budget was then used to calculate an estimate for the maximum runtime for our device based on the power consumption of each component. Once this estimate was found, testing ensued. Testing consisted of individual parts testing, subsystem prototype testing, and full system testing. From testing, we were able to conclude that our system would realistically last as long as the client desired.

3.1.2 Component Research, Selection, and Testing:

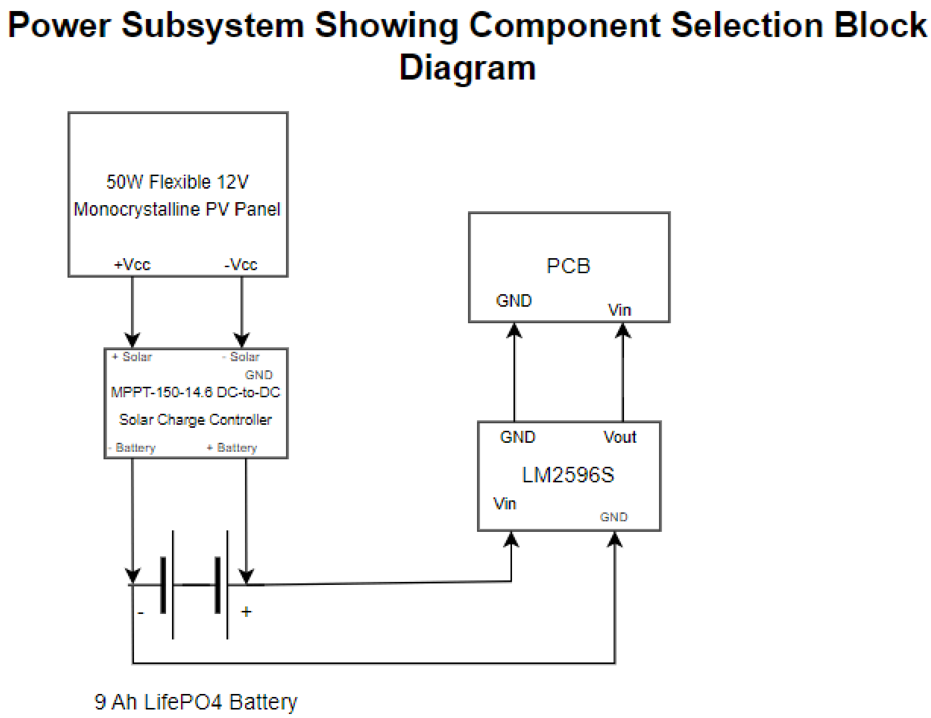

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Figure 3.1.2a: Simplified block diagram showing power components selected

The components shown in **Fig 3.1.2a** above were selected to fit the client's requirements for the device. An explanation of how the power system functionally operates can be found in section 2.2. The power subsystem consisted of four main components: the solar panel, charge controller, battery, and buck converter. Each component is essential to ensure our system meets the desired runtime of 7 days. In the client's preliminary design, his power system consisted of two components, a battery and a buck converter. The addition of a solar panel and charge controller makes it possible for the system to run well over the desired runtime, as test results will show.

<u>Component Selection: Solar Panels</u>

Components were each selected based on a list of criteria specified to fit our client's needs. The solar panel was selected based on its weight, durability, price, efficiency, availability, and size. The three types of solar panels considered were Monocrystalline, Polycrystalline, and Passivated Emitter and Rear cell (PERC) panels.

Monocrystalline Panels:

Monocrystalline panels are made from a single pure silicon crystal that is cut into multiple wafers. There are multiple advantages to this panel type. For one, they are highly efficient. Monocrystalline panels are known for their high-efficiency rates, typically ranging from 15% to 22%. They require less space compared to polycrystalline panels for the same power output, making them suitable for installations with limited space. Monocrystalline panels also tend to have a longer lifespan (often 25 years or more) due to the high-quality silicon used in manufacturing. One issue with these panels is that they are generally more expensive than polycrystalline panels due to the higher purity of silicon and the more complex manufacturing process.

Polycrystalline Panels:

Polycrystalline panels are composed of multiple silicon crystals rather than one. One benefit of these panels is that they are typically more cost-effective than monocrystalline panels. However, polycrystalline panels have lower efficiency rates (typically between 13% to 16%) compared to monocrystalline panels. On top of this, they require more space than monocrystalline panels for the same power output.

PERC Panels:

PERC panels improve on the traditional monocrystalline panel, providing more efficiency by reflecting light back into the cell. One benefit of this is that they have a higher efficiency than traditional monocrystalline and polycrystalline panels, typically in the range of 18% to 23%. The rear-side passivation technology reduces electron recombination losses, improving overall efficiency. They also perform better than standard panels in low light conditions, enhancing energy production in cloudy or partially shaded environments. The durability and lifespan of these panels are also improved. While these are great panels, one major issue is that PERC panels are significantly more expensive than traditional monocrystalline and polycrystalline panels due to the additional manufacturing complexity.

For this project, we opted to go with a flexible monocrystalline solar panel. This panel was ideal for the project, as they are highly efficient, extremely durable, portable, and available to ship to Kenya. The specific solar panel that was selected for this project was the Renogy Monocrystalline Flexible Solar Panel 50W 12V. The panel was chosen for being fairly cheap ($80), having a maximum power output of 50W, and being extremely durable. The Panel also follows proper engineering standards, having an Ingress Protection rating (IP rating) of 168. According to the standard, an IP rating of 168 means that the solar panel is water resistant in fresh water to a maximum depth of 1.5 meters for up to 30 meters [3]. This is more than adequate for rainy conditions in Kenya. It is important to note that this panel also follows IEEE Standard Definitions of Terms For Solar Cells, a standard that provides acceptable terms for use in the application of solar cells to power systems [15].

Component Testing: Solar Panel (Renogy Monocrystalline Flexible Solar Panel 50W 12V)

Below are the specifications for our solar panel:
- Current short circuit: 2.92
- Current optimal: 2.71
- Voltage open circuit: 22.6
- Voltage optimal: 18.5

To ensure that our solar panel operated as expected, we had to test it. To test to see if a solar panel is operating correctly, the short circuit (I_{SC}) current and open circuit voltage (V_{OC}) must be measured. To test V_{OC}, we connect the positive lead of the multimeter to the positive wire of the solar panel, and the negative lead of the multimeter to the negative wire of the solar panel (in direct sunlight). To test I_{SC}, we do the same thing, with our multimeter lead connected to the current port. The measured V_{OC} was 22.6, while the measured I_{SC} was 2.92. This is as expected, meaning our panel operates as expected.

With this information, we can calculate the maximum power point of this panel. The maximum power point is the point on a voltage versus current graph that provides the maximum power output of a solar panel. For this solar panel, the maximum powerpoint would be the optimal current times the optimal voltage. MPP for the panel: 2.71 A * 18.5 V = 50.135 W.

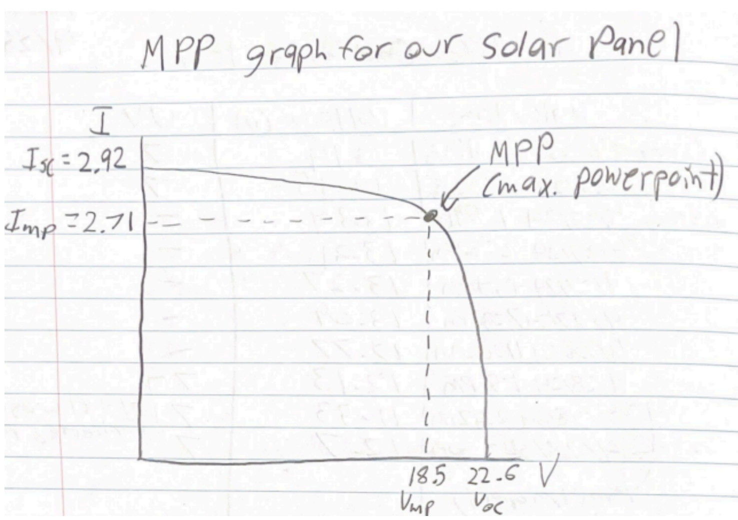


Figure 3.1.2b: Graph showing calculated max power point of solar panel used in the project.

Component Selection: Battery

The battery was selected based on efficiency, durability, weight, price, and availability. Because it is difficult to take batteries overseas, it was essential that an equivalent to the battery we used for this project could be purchased in Kenya for our client. Because of this, we were limited to two types of batteries, Lead Acid and Lithium Ion.

Lead Acid Battery:

A lead acid battery is a battery that has lead plates submerged in sulfuric acid electrolyte. Lead acid batteries are known for being cost-effective. While they are a cost-effective battery, they have a low life cycle when compared to Lithium-Ion batteries, especially when deep cycling of the batteries is involved.

Lithium-Ion Battery:

Lithium-ion batteries are batteries where lithium ions move from the negative electrode to the positive electrode during discharge and back during charging. Lithium-ion batteries have a higher energy density than lead-acid batteries, providing more energy storage capacity for their size and weight. They also have a longer cycle life and can withstand a higher number of charge-discharge cycles compared to lead-acid batteries, especially when not deeply discharged. Lithium-ion batteries are lightweight and compact, making them ideal for applications where weight and space are critical factors. On top of this, they require minimal maintenance compared to lead acid batteries, typically only requiring periodic charging and monitoring. Some issues to consider with lithium batteries are the cost and the safety concerns. Lithium-ion batteries are more expensive than Lead-acid and can be more sensitive to overcharging, deep discharging, and temperature extremes. This is why most lithium-ion batteries have built-in protection circuits to manage charging and discharging safely.

For our project, we decided to go with a Lithium-ion battery. The long cycle life, higher energy density, and portability (lightweight) outweigh the extra price. The battery we decided to go with was the LiFePO4 12V 9Ah deep cycle lithium battery by HWE. This battery was selected for having an equivalent in Kenya and meeting the standards described in IEEE Std 1679.1. This standard is a guide for evaluating a lithium-ion battery in stationary applications. One important section that was looked at from this standard was section 5.5: Operating conditions [14]. It was essential that our battery was rated for the temperatures in Kenya during July, August, and September. This battery had an optimal operating temperature range of 14 degrees Fahrenheit to 112 degrees Fahrenheit. The maximum temperature reached in Kenya during these months is around 80 degrees Fahrenheit, meaning the batteries should operate without issue.

<u>Component Testing: Battery (LiFePO4 12V 9Ah deep cycle lithium battery)</u>

Below are the discharge curves for both the battery we used and the batteries available in Kenya. Discharge curves are graphical representations that show how a battery's voltage changes over time as it discharges. These curves are crucial for understanding and predicting the performance of batteries in various applications.

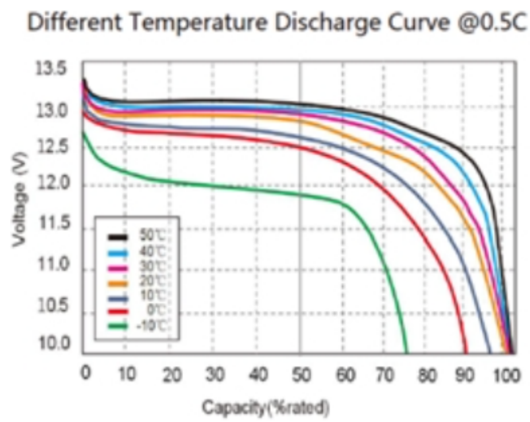


Figure 3.1.2c: Discharge curve of our in the US

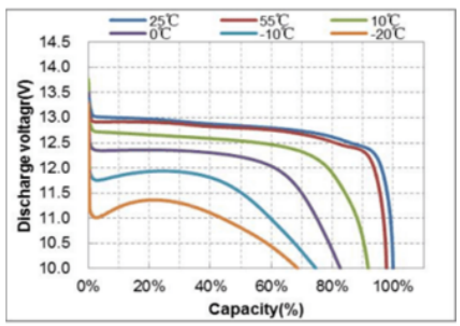


Figure 3.1.2d: Discharge curve of Mustek battery (batteries to be used in Kenya)

As can be seen in **figures 3.1.2c and 3.1.2d,** the battery discharge curve is fairly similar when comparing our battery to the ones used in Kenya. This means that the battery can realistically be used as an accurate representation of how the Kenya batteries operate.

Depth of Discharge to Cycles (DoD)

DOD	CYCLES
30%	8000
50%	5000
80%	3000
100%	2000

Figure 3.1.2e: Number of cycles in lifetime in relation to the depth of discharge drained from the battery

Figure 3.1.2e shows us how depth of discharge (DoD) affects the total number of cycles for our battery. DoD refers to the extent to which a battery's stored energy is used before it is recharged. As can be seen in the chart, if our LiFePO4 battery is drained to 100% DoD consistently, we will be able to obtain a maximum of 2000 cycles from the battery. However, if the battery is not drained to 100% DoD, it can be cycled significantly more. It is important to note that the DoD is not the same as maximum capacity. Our battery has a microchip inside that prevents it from discharging to 100%. 100% DoD is equivalent to 80% of the battery's full capacity. From **Figure 3.1.2c and Figure 3.1.2d**, we can see that the max voltage (full capacity) of the battery is 13.3V, while 0% capacity is 10V. To test this, I fully charged the battery and checked the voltage. I then completely drained the battery to see what the minimum voltage I could achieve was. The minimum voltage allowed by the battery was 11V (100% DOD, 80% of capacity), while the maximum was 13.3V. This confirmed that our battery was functional and operated as expected.

<u>Component Selection: Charge Controller</u>

For charge controllers, the main criteria considered were efficiency and price. The two types of charge controllers considered for this project were Pulse Width Modulation charge controllers (PWM), and Maximum Power Point Tracking (MPPT) charge controllers. Understanding how these two charge controllers work was crucial in the selection of the component.

PWM Charge Controller:

A PWM charge controller regulates the charging of the battery by gradually reducing the amount of power delivered to the battery as it reaches full charge. It maintains a constant voltage for the battery (typically 12V or 24V) and adjusts the charging current based on the battery's state of charge. They are moderately efficient but do not compare to an MPPT charge controller. They can efficiently charge batteries that are not deeply discharged but may struggle with low battery conditions or mismatched panel and battery voltages. These charge controllers are suitable for small solar systems or where cost is a significant factor. They are straightforward and reliable but lack the efficiency optimization of MPPT controllers.

MPPT Charge Controller:

An MPPT charge controller dynamically adjusts the voltage and current to extract maximum power from the solar panels under varying sunlight conditions. It constantly tracks the maximum power point of the solar panel array and adjusts the output to match the battery voltage, maximizing the efficiency of the solar system. An MPPT charge controller is highly efficient. It can significantly increase the charging efficiency (typically 95% or higher) by matching the solar panel's output to the battery's optimal voltage.

For this project's purposes, a highly efficient charge controller was necessary to ensure our device would last a total of 7 days. Because of this, we opted to go with an MPPT charge controller. The specific charge controller we went for on this project was the Powerwerx MPPT-150-14.6, DC-to-DC Solar Charger Controller. This charge controller was ideal for our project, as it is designed for lithium ion battery charging and was cheap for an MPPT charge controller ($30). The charge controller also follows proper engineering standards, and has an IP rating of IP67. This means that the charge controller is waterproof in up to one meter of water for up to 30 minutes [3].

Component Testing: Charge Controller(Powerwerx MPPT-150-14.6, DC-to-DC Solar Charger)

It was necessary for the charge controller to be tested to ensure that it was a true MPPT charge controller. To test this, we connected the solar panel, charge controller, and battery. We then put the solar panel in the sunlight, and measured the current of the panel to the charge controller. We then measured the current and voltage coming out of the charge controller going into the battery.

Results:

- 2.34 A from panel to charge controller
- 3.33 A from charge controller to battery
- 14.6V out of charge controller

From this test, we can see that our charge controller was able to increase the current coming out of the solar panel and going to the battery. This confirms that the charge controller is a true MPPT controller. Our charge controller was rated for 95% efficiency. When multiplying our output voltage by our current, we can see how much power our charge controller creates, and compare that to our solar panel's maximum power point. 14.6V * 3.33A = 48.618 W. The MPP of our panel is 50.135W. This means that our charge controller is 97% efficient, which is better than expected.

3.1.3 Power Budget:

One of the critical documents developed for this project is the power budget. The power budget comprehensively outlines the power consumption requirements for each component of our device. It includes anticipated power draw for each load, defines operational states (including on, idle, and off), and specifies the duration spent in each state. The power budget is essential as it enables us to accurately estimate the device's operation time. Below is our power budget:

Item	Value	Quantity	Nominal Voltage (V)	Needed Input V Regulation	Output V Variation	Spectral Purity/Noise
Battery (Source)	9Ah LiFePo4 Mustek	1	12.8 output	0-14.6	10-13.6	N/A
Charge Controller	Powerwerx MPPT-150-14.6, DC-to-DC	1	14.6	16-25	14.6	TBD
Solar Panel (Source)	SN-PH40W	1	N/A	N/A	18.5-22.6	N/A
Microcontroller	ESP32-S3 WROOM	1	3.3	3.0-3.6	2.64 - .33	N/A
MP3 Player + Speaker	DFPlayer Mini	1	3.3	3.3 - 5	N/A	TBD
DC-DC converter	LM2596S	1	12.8	3.0 - 40	1.5 - 35	50mV switching noise

State Based Current (mA)			Time in State (hrs) = 24			Total Consumption (mA-hr)/day
High	Low	Off	High	Low	Off	
N/A	N/A	N/A	N/A	N/A	N/A	N/A
N/A	N/A	N/A	N/A	N/A	N/A	N/A
2710	N/A	N/A	N/A	N/A	N/A	N/A
160	0.24	0	10	14	0	1603.36
380	20	0	10	14	0	4080
3000	TBD	0.08	10	14	0	1534.5072

Max Power (mW)	Min Power (mW)	Total Energy/day (mW-hr)	Design Notes:
N/A	N/A	N/A	9Ah * 12.8V =115.2 Wh battery cap
113.157	4.241292	N/A	95% efficiency, the power shown here will not affect the total power consumption, as it takes place before the battery (from solar panel), however this will be accounted for for thermals
N/A	N/A	N/A	Solar panel is rated at 50W (wnen directly facing the sun, best case), with an average of 12 hours of sunlight in Kenya per day.
528	0.792	5291.088	
1254	66	13464	MP3 player is running at 3.3 V, max power is 3.3V * 150A (max current). Runs for 10 hrs a day, we are given 4.95Wh/day. (66mW (idle) - 495 mW (running))
481.14	18.03384	5063.87376	73% efficiency
Total Power Consumption (Max):	**Total Power Consumption (min):**	**Total Energy/day (mW-hr):**	
2263.14	84.82584	23818.96176	

Total Running hours:	At a ~40% duty cycle, we are looking at around 23.818 W-hrs used a day. Our battery has a 92.16 Wh capacity. 92.16/ 23.818 W-hr/day = 3.7 days * 24hr = 92.86 hrs
92.86047067	

Fig 3.1.3a: Final Power Budget

From the power budget, we were able to calculate the maximum amount of time our device could operate for in the field. The total amount of running time calculated for our device at a 40% duty cycle was 92.86 hours, which equates to just under 4 days. This is a worst case scenario, assuming our device runs at maximum power constantly. This also does not account for the solar panel.

3.1.4 Subsystem Testing:

Testing took place with a 5 ohm power resistor attached to our DC-DC converter connected to our battery. The aim of this was to mimic our maximum power consumption for the device (2.263 W). Voltage was measured at the battery every couple hours to obtain accurate discharge time.

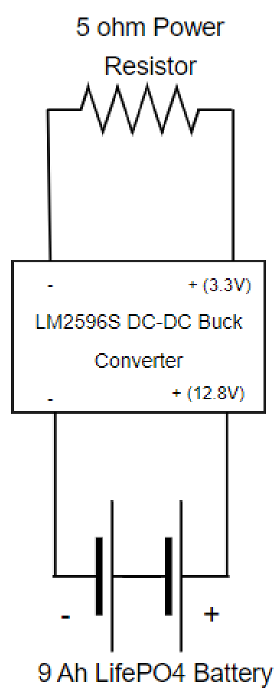


Figure 3.1.4a: Battery test circuit schematic.

One problem with this test was the fact that the DC-DC buck converter's power consumption was also taken into account when choosing our power resistor. As seen in **Figure 3.1.4a**, the DC-DC converter is part of the system, meaning that the power consumption should not have been accounted for. To solve this, we factored out the power consumption of the DC-DC converter, giving us a more accurate power consumption of 1.8 W. For a power consumption of 1.8 W, we would want to use a 6 ohm resistor. 5ohm/6ohm = .833, meaning that our actual system should be 17% more efficient than our test. To calculate the estimated runtime for our test, we took the usable power of our battery and divided it by the power consumption. 92Wh/2.2W = 40.7 hrs. We then accounted for the efficiency of our buck converter (73%), giving us a targeted runtime of 29.7 hours for this test. When accounting for the 17% extra efficiency we should realistically be getting, our estimated battery runtime comes to 37.31 at full load. Two subsystem tests were performed. Below are the results:

Subsystem Test 1

Date/Time	Voltage (v)
4/10/24 7:25 pm	13.2 - start
4/10/24 9 pm	13.13
4/10/24 10:24 pm	13.11 - stopped
4/11/24 11:28 am	13.13 - start
4/11/24 2:44 pm	13.05
4/11/24 6:24 pm	12.87 - stopped
4/14/24 4:30 pm	12.94 - start
4/14/24 6:32 pm	12.72
4/14/24 9:30 pm	dead

Figure 3.1.4b: Battery draining test with power resistor (5ohm)

Subsystem Test 2

Date/Time	Voltage (v)
4/23/24 5:23 PM	13.2 - start
4/23/24 6:23 PM	13.09V
4/23/24 6:55 PM	13.08 - stopped
4/23/24 8:55 PM	13.13 - start
4/23/24 10:43 PM	13.06
4/23/24 11:30 PM	13.06 - stopped
4/24/24 2:38 AM	13.11 - start
4/24/24 2:14 PM	12.75 - stopped
4/24/24 4:14 PM	12.83 - start
4/24/24 5:09 PM	12.7
4/24/24 5:49 PM	12.67
4/24/24 7:32 PM	12.5
4/24/24 8:34 PM	11.95
4/24/24 9:10 PM	11.04
4/24/24 9:21 PM	dead

Figure 3.1.4c: Battery draining test with power resistor (5ohm)

The results from Subsystem test 1 were invalidated, as the battery abruptly died. After recharging the battery, it was still unable to discharge or show a voltage reading when using a multimeter. A few days later, the batter began to work again. We assumed that the circuit inside the battery was a little messed up, and subsequently used a new battery for Subsystem test 2. Subsystem test 2 ran as anticipated. The battery lasted a total of 24.57 hours, with our estimated goal being 29.71. This means that we were off by 18% from our estimated goal for this experiment. Calculating how long our battery would last in a realistic environment (6-ohm resistor instead of 5 ohm), our battery would have lasted 30.59 hours. While these numbers seem low, these estimates assume a 100% duty cycle with the absolute maximum power consumption for our system.

The first solar test conducted took place with the panel, battery, and charge controller. No load was applied. The aim of this test was to observe how long it would take for the battery to fully charge. The test took place on a sunny day, with a UV rating of 7. Below are the results.

Charging a dead battery with no load

Date/time	Voltage (V)	UV
4/27/24 11:47 AM	11.04 - start	7
4/27/24 12:50 PM	12.82	7
4/27/24 1:52 PM	13.45 - stop	7

Figure 3.1.4d: Battery charged with UV 7, no load

From this test, we can see that the solar panel was able to successfully charge the battery in 2 hours. Based on this, we can see that as long as it is a sunny day, the panel is able to charge the battery from fully drained to full charge with no load extremely quickly.

The next solar test consisted of the battery, charge controller, solar panel, the DC-DC converter, and the 5 ohm power resistor load. The aim of this test was to observe whether the solar panel would charge the battery even while under full load. Two tests were performed, one on a fairly sunny day with a UV rating of 6, and another on a less sunny day with a UV rating of 4.

Charging Battery with load Test 1

Date/time	Voltage (V)	UV
4/28/24 11:47 AM	12.13 - start	6
4/28/24 12:50 PM	12.7 - stop	6

Figure 3.1.4e: Battery charged with UV 6

Charging battery with load Test 2

Date/time	Voltage (V)	UV
5/10/24 11:47 AM	12.10 - start	4
5/10/24 12:50 PM	12.3 - stop	4

Figure 3.1.4f: Battery charged with UV 4

From these two tests, we can conclude that the battery is able to be charged under load on days where the UV is 4 or higher. The average UV rating in Kenya in July, August, and September is 5. This means that on an average day in Kenya, the battery would be able to be successfully recharged.

3.1.5 Full System Testing:

Once our system was fully built, we were able to run a full system test to see how long the battery could run the final device for. For this test, we connected the battery to our device and played our human noises track. The reason we ran the human noises track was because they consume the most power when compared to the other tracks. The test was run at a 60% duty cycle, without the solar panel attached. Below are the results:

Battery Test at 60% Duty Cycle With Human Noises

DATE	BATTERY VOLTAGE	TIME
6/5/2024	13.20V	7:30 AM
6/5/2024	13.18V	12:00 PM
6/5/2024	13.15V	5:30 PM
6/5/2024	13.10V	8:00 PM
6/6/2024	13.06V	7:30 AM
6/6/2024	13.05V	12:00 PM
6/6/2024	13.01V	5:30 PM
6/6/2024	12.93V	8:00 PM
6/7/2024	12.88V	7:30 AM
6/7/2024	12.85V	12:00 PM
6/7/2024	12.80V	5:30 PM
6/7/2024	12.71V	8:00 PM
6/8/2024	12.63V	7:30 AM
6/8/2024	12.60V	12:00 PM
6/8/2024	12.56V	5:30 PM
6/8/2024	12.46V	8:00 PM
6/9/2024	12.33V	7:30 AM
6/9/2024	12.30V	12:00 PM
6/9/2024	12.28V	5:30 PM
6/9/2024	12.23V	8:00 PM
6/10/2024	12.11V	7:30 AM
6/10/2024	12.08V	12:00 PM
6/10/2024	12.03V	5:30 PM
6/10/2024	11.98V	8:00 PM

Figure 3.1.5a: Final system testing shows system lasts multiple days

From this test, we were able to determine that without a solar panel, the device was able to last for 6 days. From the tests conducted, we can conclude that the power systems engineer was successful in providing the client with the desired deliverables. While the full system test lasted 6 days instead of 7, this was without a solar panel attached to the device, and a higher duty cycle than desired by the client (60% instead of 40%). With the correct duty cycle and a solar panel attached, the device would easily last 7 days. As stated previously, the average UV rating in Kenya is around 5 during July, August, and September. As long as the device gets one day with a UV rating equal to or greater than 4 in the 6 days it is running, the battery will be able to fully recharge.

3.2 Embedded Systems

For a project requiring sophisticated components found in an embedded system, the composition of certain parts contributed significantly to the backbone that allowed our group to satisfy the client deliverables given to us in the pre-design phase of this device. To achieve this, the team chose to design and integrate a specialized computing system to perform dedicated functions within the entire electrical system of the playback device. Tailored to the group's client, the embedded system was designed to perform specific tasks and functions that would ultimately contribute to the reliability and functionality of the project. This would help progress our client's main goals of wildlife conservation and provide peace of mind that their unmonitored device is performing as intended. Defined as a subsystem, the embedded system features a central brain that acts as the center of functionality in the device. Using this design, the embedded systems engineer can leverage this centralized brain to execute carefully written programmed instructions that drive anything from I/O interfaces to internal functionality. With this seamless hardware and software integration, the embedded system operates efficiently and effectively to fulfill its specific role within the larger device. To accomplish this, the subsystem itself must be divided into smaller, equally important components that give us an overview of how our brain works from the inside out.

3.2.1 Subsystem Overview

The embedded system of the wildlife playback device would not be able to fulfill its goal of satisfying client preferences without the use of the centralized microcontroller used in the project. Ultimately, the microcontroller coordinates and manages the various components and functions to ensure seamless operation within the device. Derived from the client preferences, the microcontroller has an important job in helping our client get the best out of the device. External buttons programmed through it allow the user to interact with and control the device. UART communication within the board facilitates interfacing with the MP3 module that stores music, sending commands so that audio playback and control are at the user's fingertips. Internal sleep functions enable the controller to conserve power when the user is not changing parameters, ensuring only essential peripherals remain active. Lastly, the analog-to-digital converter capability allows battery monitoring and system feedback to be read out in the field and analyzed for future use. Translating these functions into what the client wants takes time and consideration in a pre-design phase that ensures the components selected are capable of executing these abilities and are compatible with the surrounding subsystems in the device.

To perform this pre-design phase correctly and save time and effort in the future, it's essential to have a proper understanding of what the client wants, what the client needs, and how those two things translate into the technical world of engineering and the embedded system. For the group's client, a few key needs stated require extensive requirements for both microcontrollers and the surrounding components that comprise the embedded system. In summary, the client's needs relating to the embedded system are as follows:

- **External controls**: The device must be accessible and functional outside of changing code. To achieve this, adding external control through the use of buttons allows for changing volume and treatments on the go and out in the field.
- **Maximum power saving abilities**: The device must be as power efficient as possible. Therefore, the ability to turn off power-hungry peripherals and enter power-saving modes when not essential can be the difference between the device lasting hours versus days.
- **Monitoring capabilities**: The device should be able to reflect the condition of the main battery powering the system. This means utilizing a microcontroller's ability to convert analog readings into digital data that can be read and processed.

After defining these client requirements, the group was able to add their design requirements that would contribute to the technical integration of the embedded system into the rest of the device and its subsystems. The design requirements for the microcontroller decided by the group are as follows:

- **3.3V power requirements**: To ensure compatibility with other subsystems and maintain some level of universality within the entire device, the microcontroller should follow a common operating voltage. This allows the embedded system engineer to integrate it more easily with other subsystems using similar devices with similar operating voltages.
- **UART communication capabilities**: Similar to the power requirement, the microcontroller should be capable of communicating effectively with other chips or boards on the device. A common method of this is the communication protocol called Universal Asynchronous Transmitter/Receiver (UART), which provides a low-energy way to send and receive commands between devices.
- **Size requirements**: The device as a whole needs to be compact and designed to be carried in mass quantities by the client. One way to help with this requirement is to use small microcontrollers that fit compactly without sacrificing the computing power of a regular-sized microcontroller.

After laying these design requirements out, the microcontroller selected for the embedded system brain is the **ESP32-S3-DevKitC-1-V1.1** by ESPRESSIF. Below is an image of the microcontroller along with its pinout diagram and some of its peripheral capabilities:

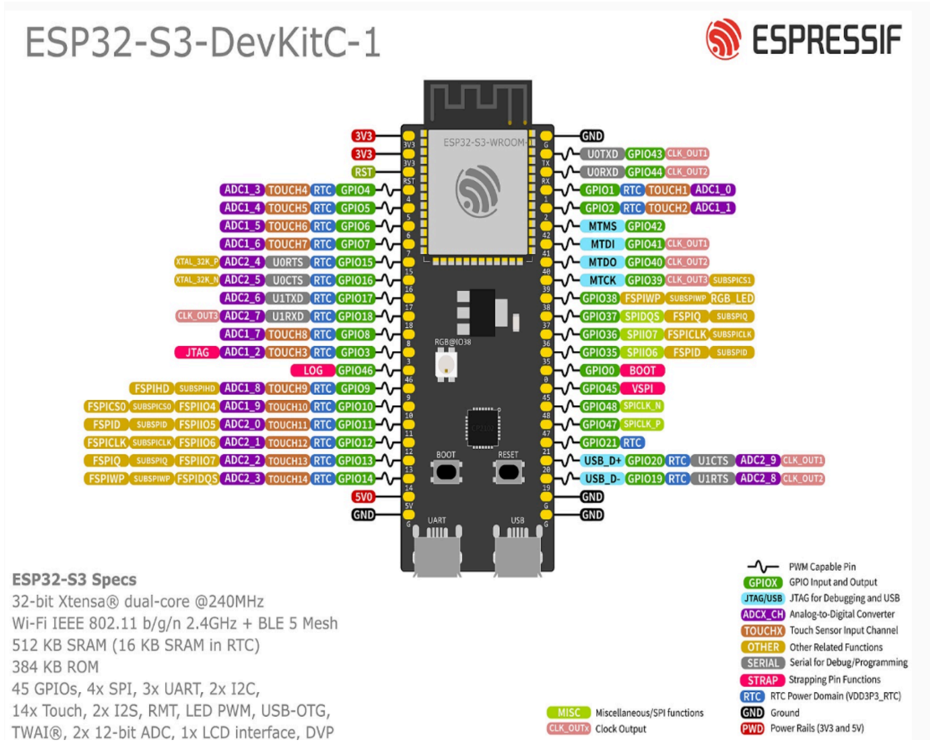


Figure 3.2.1.a - ESP32-S3 Pinout and Peripheral Diagram [10]

The ESP32-S3 microcontroller is a versatile and powerful component well-suited for our embedded system. It can read inputs from the external buttons through its numerous GPIO pins, enabling the ability to integrate user interaction and control. This capability proves to be powerful when user interaction is then routed from the buttons to the board's ability to support UART communication through its Rx and Tx pins. This is the key to facilitating reliable data exchange with the MP3. In addition, the multiple ADC pins allow a large selection of where to place the external voltage reading which will then allow a programmed instruction to turn this analog signal into digital data for precise measurement and processing. With such a powerful brain ready to be integrated into the embedded system, subsystem circuit prototyping commenced with the goal of putting the technical decisions from the pre-design phase to the test.

With the help of Aiden and his ability to create schematics in Cadence, the embedded system was assembled with the essential components to help accomplish a functional embedded subsystem. Below is a schematic diagram of the prototype schematic with the highlighted embedded system circuit in red.

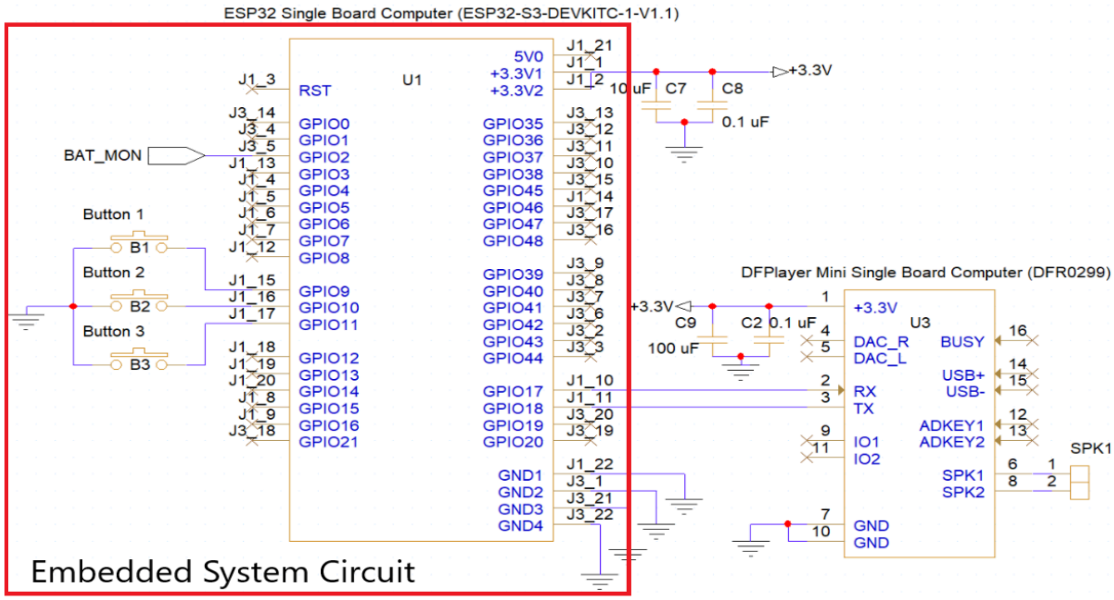


Figure 3.2.1.b - Prototype Schematic with Highlighted Embedded System Circuit

The wiring diagram illustrates the connections for the embedded system utilizing the ESP32-S3 talked about previously. In addition, the audio subsystem, located to the right of the red outline, is depicted to help visualize how the two are connected when talking about communication between subsystems. Serving as the central unit of the circuit, the ESP32 Single Board Computer is powered by the 3.3V supply connected at the top. In addition, to the right of the board shows where the device's three external buttons reside. Using a common ground connection, the three external buttons utilize the microcontroller's capabilities to interface with GPIO pins 9, 10, and 11. Just above these pins is where the input for the ADC pin reads the battery monitoring circuit in GPIO pin 2. As mentioned, the MP3 module shown above shows how the DFPlayer Mini and the ESP32-S3 achieve simple UART communication with the connection of 4 pins. The Rx pin of the DFPlayer (pin 2) connected to the Tx pin of the ESP32 (pin 17) allows for the sending of communication from the controller to the MP3. Similarly, the Tx pin of the DFPlayer (pin 3) is connected to the Rx pin of the ESP32 (pin18), allowing for the return of information back to the board from the MP3. This configuration of pins and connections allows for the ESP32 to effectively manage external inputs, monitor battery voltage, and facilitate audio playback through UART communication with the DFPlayer mini. After a preliminary design of the embedded system is done, the same must be done for the software architecture that helps create functionality out of what was just designed.

3.2.2 Technical Details

Designing software architecture can be just as difficult as designing the circuit for the embedded system and should follow similar steps of design to ensure it is done correctly. Beginning the software architecture before beginning to even write the code for the project is crucial for several reasons. First, it provides a clear blueprint for, defining how different components interact and ensures that all requirements are met efficiently. This upfront planning helps to identify potential issues and complexities early, reducing the risk of significant rework later in the development process. Meaning, that a well-thought-out architecture enhances maintainability and scalability, making it easier to adapt the system to future changes or expansion like the animals our client hopes to save from this project. For the group's project specifically, preparing a software architecture meant using a structured approach to figuring out how communication protocols, data handling, and I/O integration could all be integrated into one perfect design.

For communication protocols, defining the UART configuration meant the difference between hours of troubleshooting and the device working right away. Baud rate, data bits, parity, and stop bits ensure that the devices connect properly, communicate clearly, and maintain connection throughout functionality of the device. For our MP3, the module has a UART port with standard serial and an adjustable baud rate (Default baud rate is 9600) [9]. Using the microcontrollers' similar adjustable baud rate capability, the MP3 and ESP32 can be linked up to a similar baud rate while leaving communication with the computer for troubleshooting. Using simple code manipulation and the serial pins of the board, the two are synced up and sharing data at the same rate, ready to begin sending information, commands, and data back and forth between one another. Once the communication is squared away, determining how to send commands with external buttons can be considered next.

For I/O integration in this software architecture, handling of the external buttons must be considered in detail and mapping each button to specific GPIO pins helps expedite the process. Developing interrupt service routines and polling mechanisms to detect button presses while at the same time, considering the need to debounce buttons to ensure reliable input. This consideration translates into how well the communication protocol can pass it on to the MP3. A large part of this, button debouncing, will be talked about in the challenges and solutions portions of this report along with the design considerations associated with it. With information like this being crucial to how the device performs, it becomes clear that handling data like this becomes of utmost importance when collecting inputs from external factors such as buttons and batteries.

The data handling framework, specifically for incorporating ADC pins to read external voltages, requires a specification for how the conversion from analog to digital is to be managed. Sampling rates, resolution, and voltage reference can all contribute to difficulties if not properly planned out. Done right the battery monitoring for the device becomes accurate and useful to both the team and the client. For the embedded system design, the sampling rate does not have to be something that is consistently taking voltages. Instead, we can rely on the slow depleting nature of the battery to lengthen the time between sampling rates. Something along the lines of 4 samples a day would work well for our device. An important factor for the ESP32 is the ADC resolution it can provide. The data sheet for the ESP32 reveals a 12-bit ability at each of the ADC pins on the microcontroller [11]. Last and certainly the most important for the device is the reference voltage which comes in at 3.3V. This will be another topic discussed in the challenges and solutions section of this report to explain the design that went into considering this factor.

Many of the topics listed above demonstrate a basic understanding of what goes into the software architecture. However, when implementing these topics into the real design of the embedded system circuit, technical challenges arise that demonstrate the importance of good engineering standards in designing software architecture. More specifically, how consideration for button debouncing and reference voltage at the ADC pin heavily influence design considerations in both the architecture and subsystem design.

3.2.3 Challenges and Solutions

The design and preparation for technical details can be a long and thorough process full of problems, redesigns, and solutions. Challenges like button debouncing and voltage reference considerations require design testing and extensive amounts of research about how to accomplish the end goal as efficiently as possible. The roles of each of these may differ related to their overall functionality within the system, however, the road to the end design of each of these smaller subsystems follows a step-by-step process derived from good engineering standards. The first challenge, related to the external buttons, focuses on the ability to debounce buttons.

Button debouncing is a critical technique used with external buttons on microcontrollers to ensure accurate and reliable input detection. When a button is pressed or released, the physical contact can cause multiple rapid changes in the signal known as "bounces". This behavior can lead to erratic and unintended behavior in the system. This behavior can be seen below:

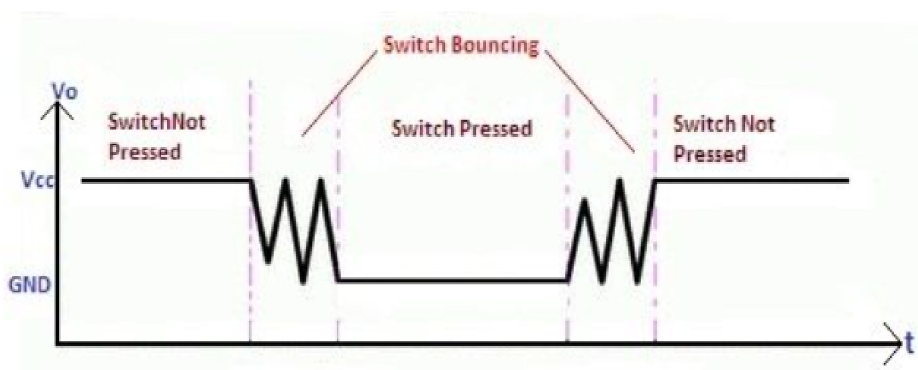


Figure 3.2.3.a - Bouncing Signal from Button [12]

The rapid up and down movement in the signal is what is referred to as the "bounce" and each transition could be mistaken as an input signal. Solutions for this can be implemented either through physical hardware debouncing or through software programming. Hardware debouncing, typically consisting of capacitors, resistors, and a logic chip, relies on creating hysteresis to smooth out the signal. On the other hand, software debouncing relies on creating short time frames to filter out the noise by filtering rapid fluctuations within that time frame.

Managing this noise can enhance the reliability and performance of the embedded system when it is added to the rest of the device. To make sure the correct design for the device was picked, it was important to test them both and make the decision afterward. Starting with the hardware solution.

Button debouncing and its effects can be effectively managed using a combination of a resistor, a capacitor, and a logic chip like the 74HC14 Schmitt trigger inverter. Below shows the first prototype of the embedded system with a display of how this circuit was to fit into our system.

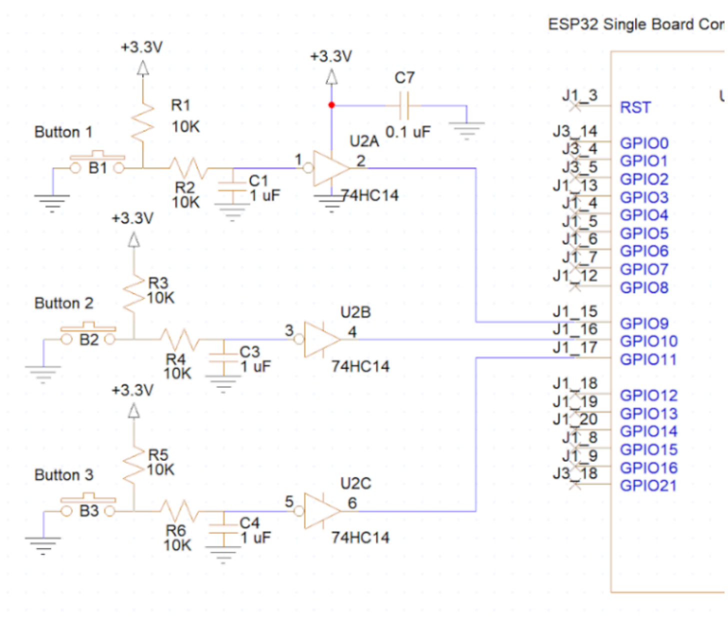


Figure 3.2.3.b - Hardware Debounce Circuit

The resistor and capacitor form an RC debouncing circuit for each GPIO pin. The resistor limits the current flow when the button is pressed, preventing damage to the components, and helps pull the input to a known voltage level when the button is not pressed. The capacitor smooths out the rapid voltage changes caused by the button's mechanical bounces by charging and discharging through the resistor, effectively filtering out high-frequency noise. The 74HC14 Schmitt trigger inverter further processes this smoothed signal before it reaches the input of the board. The Schmitt, triggered by hysteresis, switches at different voltage thresholds for rising and falling edges, thus ignoring small fluctuations and noise that might occur during bouncing [13]. This component converts the cleaned analog signal from the RC circuit into a stable digital output, ensuring the microcontroller receives an accurate representation of the button's state. Together, these components ensure reliable and predictable behavior in the embedded system by providing a clean, stable input signal free from the erratic effects of button bounce. This design, although reliable, proved to be an expensive and spacy solution to the bouncing from the buttons. As a result, a secondary solution found in the software of the microcontroller compacted the embedded system back to only its necessities.

The software solution to this problem proved to be something efficient and just as reliable. For debouncing with software, it is a common practice to implement a timer-based approach to filter out these false inputs from bouncing. Instead of the hardware smoothing it out, introducing a short delay to allow the mechanical bounce to settle before confirming the button state creates the same effect as the logic chip seen in the hardware solution. The process involves reading an initial button state and then using a timer or delay function to wait for a specific debounce period. For this device, 50 milliseconds was used. After this delay, the button state is read again and if the button state remains the same as the initial read, the button press is considered valid. Not only does this process provide the same functionality and reliability as the hardware solution, but the appealing cost and space reduction means that the software solution for debouncing the device's external buttons was a final design pick.

In addition to the button debouncing, the reference voltage detail related to the ADC pin created additional design choices that weighed heavily on power consumption and complexity. As discussed, the microcontroller utilizes an operating voltage of 3.3V at the power pins. As a result, the other pins used for I/O capabilities must not exceed this voltage in order to protect the microcontroller. The reason for this is that the ADC pin can only accurately measure up to what the microcontroller's supply voltage is. Defining these points along the supply voltage (0V to 3.3V) is what allows the ADC to accurately assign numbers for the microcontroller. As a result of wanting to directly measure the 12-volt battery, it is essential to divide the maximum voltage of the battery by at least the maximum voltage of the operating voltage of the microcontroller. To do so, the implementation of a voltage divider circuit was required to perform this task.

The Voltage Divider is tasked with reducing the 12-volt battery voltage to a level suitable for precise measurement by the ADC pin (<3.3V), taking into account considerations such as current consumption, voltage ratio, ADC pin input impedance, and resistor values. Incorporating this circuitry into the ADC pin setup facilitates the management of inherent current fluctuations, irrespective of the microcontroller's operational state, including factors such as internal circuitry and input impedance. The choice of how to implement this voltage divider, however, is anything but simple. To make the correct design choice, comparing the potential ways of dividing this voltage point in the correct direction for testing this circuit. Below is the comparison of the circuits considered.

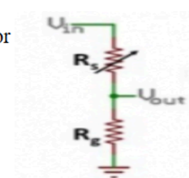
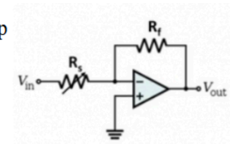
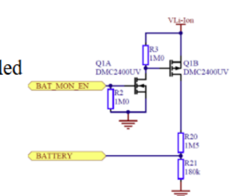
VD Circuit	Two Resistor 	Op-Amp 	MC Controlled 
Pros	• Cost effective • Simple Circuit Operation • Passive • Low Quiescent Current	• High Input Impedance • Precise Voltage Division • Buffering	• Dynamic Control • Low to Zero Quiescent Current for off state • Reduced loading effects
Cons	• Loading Effects • Limited V range • Output Impedance	• Complexity (Cost and consideration) • Power Requirements • Higher Quiescent Current	• Complexity (Cost , consideration, control) • Higher Power Consumption for On state • Noise Sensitivity

Figure 3.2.3.c - Voltage Divider Comparison Based on Project Requirements

The two-resistor method for voltage dividers presents a compelling choice compared to the op-amp and microcontroller-controlled (MC-controlled) methods due to its simplicity, cost-effectiveness, and low power consumption. First and foremost, the two-resistor configuration is highly cost-effective, utilizing only two passive components, which are inexpensive and widely available. This simplicity translates to straightforward circuit operation and easy implementation, making it an ideal choice for basic voltage division tasks without the need for complex design considerations. Another significant advantage of the two-resistor method is its passive nature, resulting in low quiescent current. Unlike the op-amp method, which requires an active component that consumes power even in idle states, the two-resistor setup does not introduce additional power requirements, thus conserving energy. This low power consumption is particularly beneficial in the device's battery-operated design. While the op-amp method offers high input impedance and precise voltage division, it comes with added complexity, higher quiescent current, and additional power requirements, which can be overkill for simple voltage division needs. Similarly, the MC-controlled method, although providing dynamic control and reduced loading effects, introduces complexity, higher on-state power consumption, and potential noise sensitivity issues, making it less ideal for this straightforward application. In summary, the two-resistor voltage divider method stands out as the best option due to its cost-effectiveness, simplicity, ease of implementation, and low power consumption, making it a practical and efficient choice for the basic voltage division requirement in our device. The final design for this can be seen below, integrated into the final prototype of the power circuit.

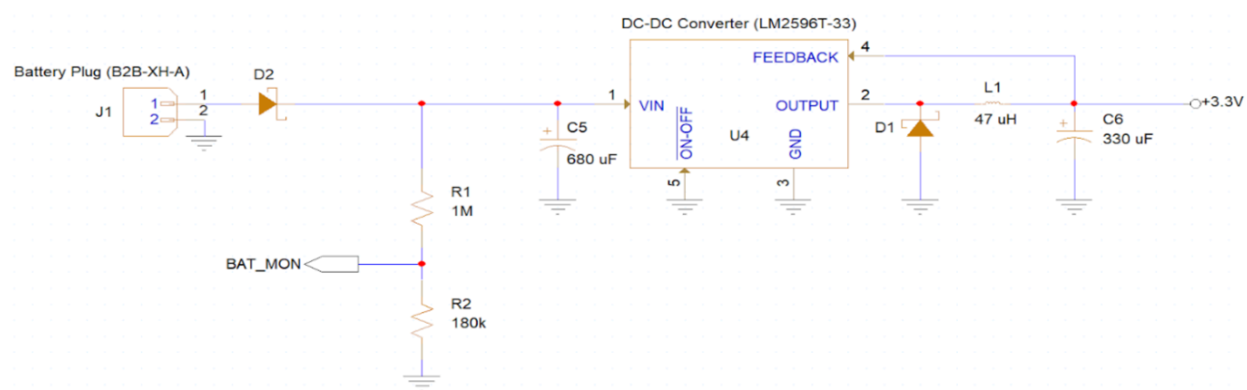


Figure 3.2.3.d - Two Resistor Voltage Divider in Final Prototype Power Circuit

This final design choice for the battery monitoring system was not the end of the design considerations for this topic. In addition to the ADC pin requiring this consideration for the reference voltage, it is also a significant design consideration when discussing the non-linear behavior of the ADC pin. The relationship of the ADC pin to the digital read can be seen below.

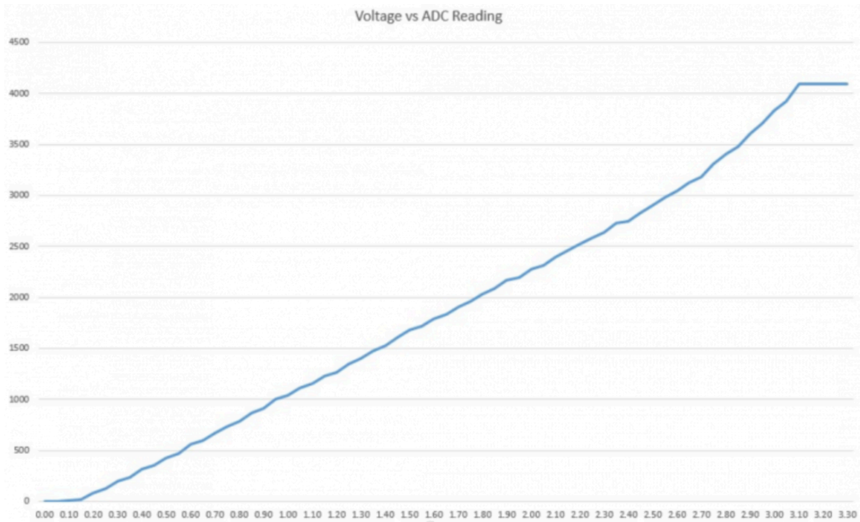


Figure 3.2.3.e - ESP32 ADC Pin Voltage Versus ADC Reading

Noticeably, the two ends of the graph show where the non-linear relationship determination comes from. To ensure the ADC pin is in fact reading what the group needs on a linear scale, it is possible to shrink the voltage that can be read to the center part of the graph. This means creating a calibration technique that instead of using a 0-3.3 scale, uses a 1.1-2.8 scale that proportionally symbolizes the voltage but on a more linear scale. This is the final design choice for this topic in the ADC pin reading.

With these design choices for external buttons and ADC pin readings, the final design for the embedded system can begin a testing and validation phase that begins to test the system in a more realistic environment. Doing this part of the process ensures the refinement of a troubleshooting routine and the confirmation of a reliable device.

3.2.4 Testing and Validation

Testing and validation are critical steps in the final development process of the device, ensuring the design meets the required specifications and functions brought to the group in the beginning by the client. Before finalizing the design, the device requires testing to help identify and rectify potential issues. This ultimately will help prevent costly mistakes and ensure the reliability of the device when out in the wild where equipment may not be readily available. This validation confirms a minimum standard that the device must be held up to in order to be ready to be sent into the wild for real-world actions. The process for testing and validation for this embedded subsystem is used to not only enhance the device's quality and safety but also build confidence in the client before their journey. For a strongly software-driven subsystem like the embedded system, testing and validation came in the form of various test benches that confirm working portions of a central, main code. These test benches are derived from the client preferences talked about in the beginning and show critical subcomponents working based on what test bench is selected for testing. Although each component had its own test bench, there were two benches that helped assist the group in troubleshooting problems during the first prototype and ultimately led to a centralized area of where the problem was occurring. As a result, changes could be made in the next design to remediate these problems.

The first of these two benches, the UART Connectivity Test, was designed to test the initial UART connection between the microcontroller and MP3 upon first powering on the device. The testbench singled out the Rx and Tx pins talked about in the technical details section of the embedded subsystem. It first tested an initial connection and if it failed, proceeded to check each individual pin to identify which connection was breaking the transmit and receive. For the first prototype, the initial connection showed connectivity issues between the microcontroller and the MP3 upon first powering it on. The group was able to then confirm the correct power supply to the MP3 and after, localize the issue to the UART communication pins. Here, the microcontroller could test the connectivity by measuring the power getting to the Rx and Tx pins of the MP3. This led us to a faulty connection in the Rx pin of the MP3, causing the connectivity to be compromised. The group identified this and remediated the problem with a new solder.

The second of these test benches, the Button Input Test, was designed to test the incoming signal from the buttons and the debounce abilities of the software. The test bench used polling methods to raise flags if problems with the buttons arise. Similar to the first test, problems arose with the buttons on the first prototype. Using the testbench, the program was showing the consistent flag being raised, indicating the button was connecting the pin to the ground constantly. This led to a redesign of the button footprint and configuration.

Although the rest of the test benches were not needed because of the successful design, the validation provided by the successful passing of these test benches ensured that the proper steps were taken for each design. In addition, client validation from these tests could be achieved.

3.3 Audio and Electronics System [Liam Magee]

3.3.1 Subsystem Introduction and Deliverables

The Audio and Electronics system is responsible for the client-specified requirements of the audio output of 80 dB SPL at 1m with client-approved sound quality. The system must be volume adjustable by a user without modification of code, the duty cycle and treatment must be in-situ user adjustable, and the device must be able to store at least 3 treatments containing 10 tracks each. As with all other subsystems, the device must withstand the environment, which includes protecting sensitive components from rain, dust, and animals, and be portable enough to be carried by one person. To achieve this rain and dust ingress protection, components selected for the audio system that will be exposed to the environment followed the IEC 60529 standard of ingress protection ratings. “The IEC has developed the ingress protection (IP) ratings, which grade the resistance of an enclosure against the intrusion of dust or liquids. The first numeral refers to the protection against solid objects and is rated on a scale from 0 (no protection) to 6 (no ingress of dust). The second numeral rates the enclosure’s protection against liquids and uses a scale from 0 (no protection) to 9 (high-pressure hot water from different angles)” [3]. The exterior components of this subsystem need to have an IP rating of IPX4 or higher to protect them from rain. A rating of IPX4 means protection from splashing of water, which is tested by spraying the device from different angles. This IP rating is suitable for rain but not submergible, which follows the requirements set by our client.

The electronics portion of this subsystem is responsible for designing the overarching network and integration of electronic components, working closely with the PCB system to achieve this. This involves ensuring that each component has the necessary power signal to achieve its required functions and that each electronic component is compatible with the rest of the system. The main designs this portion handled to achieve were the power circuit design, filtering, and hardware button debouncing before the software approach was implemented.

3.3.2 Audio and Electronics System Proposed Design

The Audio and Electronics Subsystem met the requirements listed above consisting of the following components shown below in **Figure 3.3.2.a**.

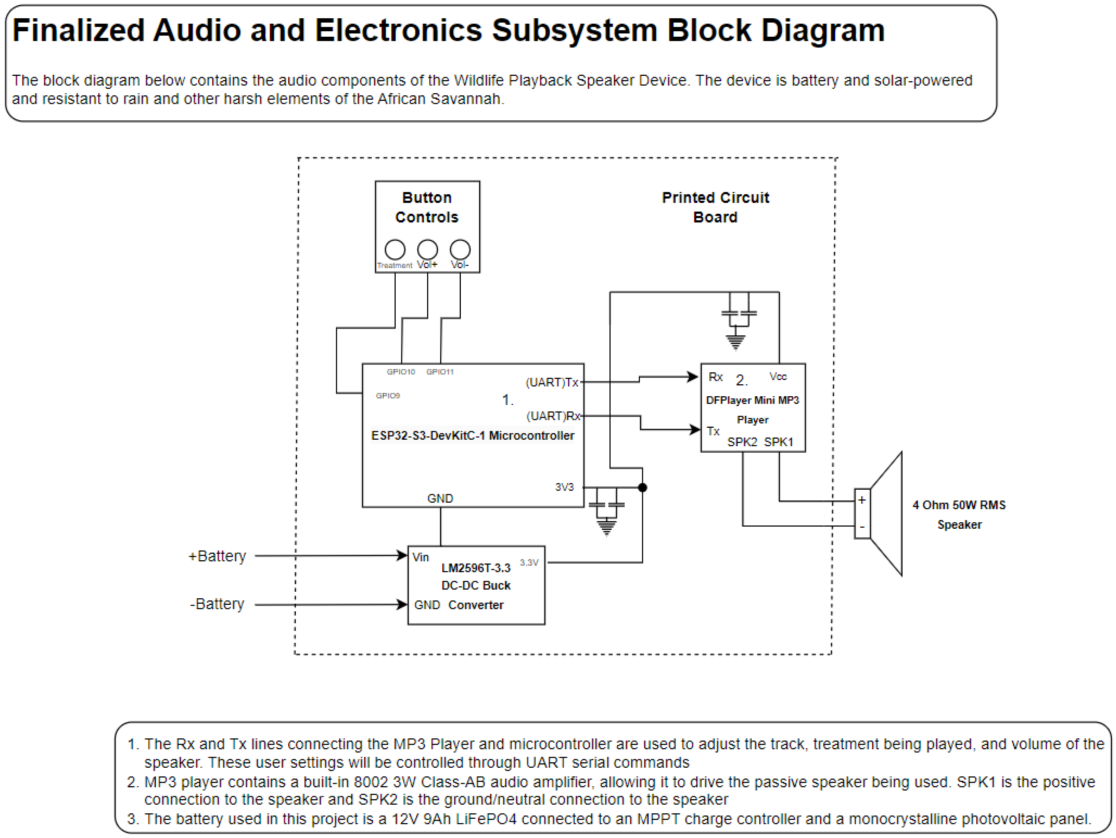


Figure 3.3.2.a: Audio and Electronics Subsystem Block Diagram, finalized after client review and feedback.

The design shown above consists of 4 main components integrated to achieve the deliverables outlined in Section 3.3.1. These components and their requirements for success based on project deliverables are:

MP3 Player Requirements for Success:

The MP3 player needs storage for at least 3 treatments containing 10 tracks each to fulfill deliverable 6 in Section 2.1 and must be compatible with a microcontroller using serial communication. Per the client's request, the MP3 must be able to play both MP3 and WAV files to allow a wider variety of track options and different levels of audio quality. The component must also have a similar operating voltage to the microcontroller (within ±3V), to reduce the size, cost, and complexity of the board for ease of client manufacturing.

Speaker Requirements for Success:

The speaker must have a frequency range between 100Hz-17kHz to support the low frequencies the cattle treatments will output and the high frequencies of the frog and human treatments. The speaker is the only component in this subsystem that will be partially outside the enclosure, exposed to the environment so it needs to conform to the IEC 60529 standards and have an IP rating of at least IPX4 to protect from ingress of water. A high speaker sensitivity of at least 85 dB SPL (1W/1m) is necessary to ensure the power efficiency of the audio system. A speaker sensitivity of 85 dB SPL (1W/1m) means that a 1W signal driving the speaker will produce an 85 dB SPL sound measured at 1m. In this project, the higher the speaker sensitivity, the better it is for power efficiency. The speaker's impedance must also allow the speaker to be driven by the amplifier being used in the device.

Audio Amplifier Requirements for Success:

Since the speaker being used in this project will likely be passive, an audio amplifier is necessary to drive the speaker. The selected audio amplifier must have a load requirement that fits in the range of the impedance of the speaker and must have a minimum output power that can drive the speaker based on its sensitivity efficiently. A low Total-Harmonic-Distortion (below 15%), is necessary to ensure the audio quality meets the client's demands.

DC-DC Step-Down Converter Requirements for Success (In Collaboration with Aiden):

The DC-DC step-down converter must have an input voltage range that covers the battery discharge characteristics (10V-14V) and an output voltage that matches the voltage required to supply the microcontroller and MP3 (3.3V). The output load current must be high enough to supply the rest of the components and the converter needs to operate at high efficiency (>70%) to help achieve the requirement of the device lasting 7 days.

To fulfill the requirements shown above, the following components were selected:

DFPlayer Mini MP3 player

The DFPlayer Mini MP3 Player is shown below in **Figure 3.3.2.b**, fulfills the requirements stated above. The component supports up to 100 folders/treatments, where every folder can hold up to 255 audio files. It accomplishes these storage requirements by having a micro SD card holder and reader that supports FAT16 and FAT32 file systems, supporting up to 32GB of storage. Using the components TX and RX pins, the DFPlayer Mini supports serial communication at a baud rate of 9600, which allows volume and treatment/folder adjustment through these serial communication ports. The component supports MP3, WAV, and WMA file formats, with sampling rates(kHz) of 8/11.025/12/16/22.05/24/32/44.1/48. The DFPlayer Mini's support for a wide range of sampling rates ensures compatibility with various audio files that may be used by the client. The component has support for a dynamic range of 90 dB and SNR support of 85 dB. A higher dynamic range means the player can handle both quiet and loud sounds without distortion and a high Signal-to-Noise Ratio indicates that the MP3 player can produce clear audio with minimal background noise.

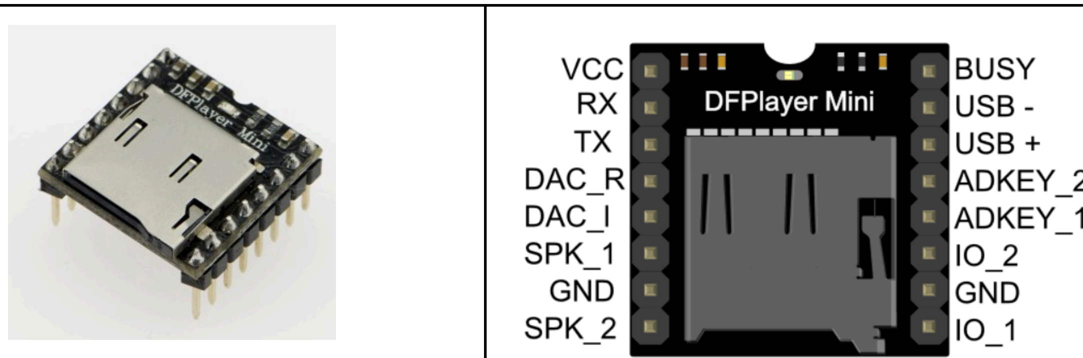


Figure 3.3.2.b: DFPlayer Mini MP3 Player

Another reason this component was selected over other MP3 player devices was its inclusion of a built-in class AB 8002 audio amplifier. This audio amplifier on the MP3 Player is capable of delivering a continuous 3W of average power output, which is more than enough to drive a speaker with a sensitivity of greater than 85 dB SPL (1W/1m) to achieve 80 dB SPL at 1m. When delivering these 3W of average power, the THD of the amplifier is below 10% and has an output load requirement of 0-8Ω.

Pyle 4in 50W RMS Marine GradeSpeaker

The speaker selected for this project, shown in **Figure 3.3.2.c**, has a sensitivity of 90 dB SPL (1W/1m), which means the speaker needs to be driven by only 0.1W to achieve the goal of 80dB SPL at 1m. The frequency response of this speaker is 90Hz - 18kHz, which covers the lower frequencies outputted in the cattle treatment and the higher frequency harmonics of the human treatment. An IP rating of IP44 is satisfactory in protecting the device from the ingress of water due to rain and larger dust particles. The 4Ω impedance of the speaker fits within the load requirements of the 8002 audio amplifier on the DFPlayer Mini.

Figure 3.3.2.c: Pyle 4” 50W RMS 4Ω Marine Grade Speaker.

Texas Instruments LM2596T-3.3 Buck Converter

A switching regulator was selected over a linear regulator for this project because switching regulators are more efficient than linear regulators and generate less heat, even though they are often noisier than linear regulators. TheTexas Instruments LM2596, shown in **Figure 3.3.2.d**, was selected for this project because of its high efficiency (73% when stepping down 12V to 3.3V), an output load current of up to 3A, and a versatile input voltage range of 0-45V.

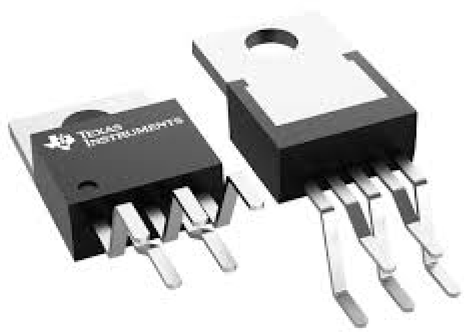


Figure 3.3.2.d: LM2596T-3.3 in the TO-220 package used in this project.

After the components were selected, the design of the PCB could begin. One of the main concerns when designing the PCB was to make sure the power supply signal integrity was sufficient enough to supply power to the microcontroller and MP3 player, allowing all deliverables related to those components to be met. This design and layout of the power circuit was done in collaboration between Liam and Aiden, as much of the signal integrity was dependent on the design of the PCB layout. The first attempt at designing the power circuit was done following the guidelines given in the LM2596 datasheet, which gave guidance for component selection and the layout, the circuit guide they followed is shown in **Figure 3.3.2.e**. The component selection tables in conjunction with the graphs given in the application information steered the group in type and values of components to use with the LM2596 to have a stable output. Aiden and Liam were then led to use application notes from Texas Instruments on switching power supply layouts by the teaching team after asking for feedback.

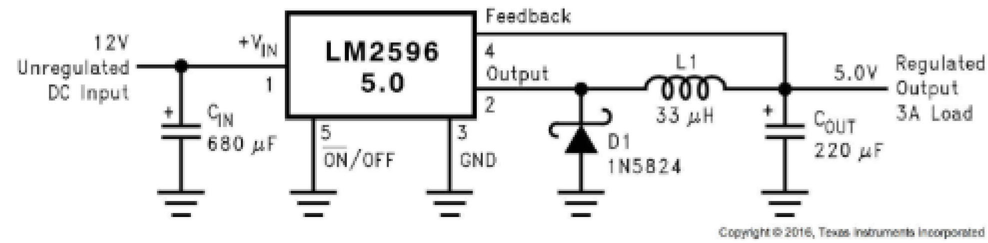


Figure 3.3.2.e: Typical application of fixed voltage LM2596 Steo-Down Regulator. Used as a guide by the team for power circuit design.

These application notes specifically AN-1229 and AN-1149 gave much more insight on how to select components and how to place them on the layout concerning the LM2596. For each extra component in the power circuit, the application notes gave suggestions for their characteristics. The inductor (L1 in **Figure 3.3.2.e**) needs to have a saturation current higher than the output load current for the inductor to function, otherwise it begins to behave more like a resistor. The inductor also must be placed close to the IC and be a low EMI inductor. The filter capacitors (C_{IN} and C_{OUT} in **Figure 3.3.2.e**) need to be low ESR and as close to the IC as possible. The catch diode (D1 in **Figure 3.3.2.e**) needs to have a fast switching speed and low forward voltage drop, making a Schottky diode the ideal selection [8]. In regards to the layout, the AC paths, are shown by the bold wires in **Figure 3.3.2.f**, need to be wide traces or copper planes and short.

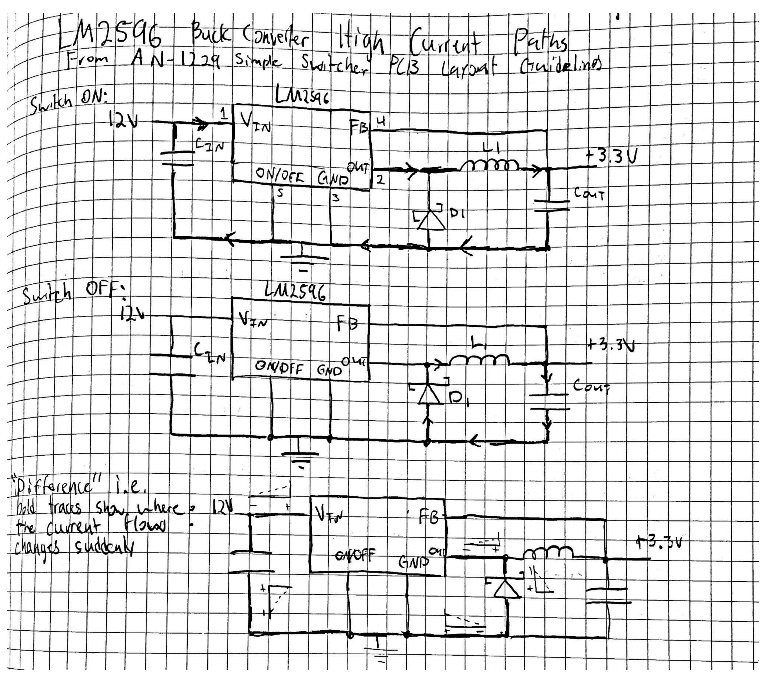


Figure 3.3.2.f: LM2596 power circuit schematic, where bold traces show where the current flow changes suddenly.

These AC paths or switching current loops also need to be arranged on the PCB such that the switching current loops switch in the same direction and the power traces should be 15mils per amp[6], [8]. **Figure 3.3.2.g** shows the layout used on the first prototype PCB, with large copper pads for AC paths and power planes for thermal management. The ground return path also needed to be considered when placing components and routing traces to avoid unnecessary ground bounce or EMI.

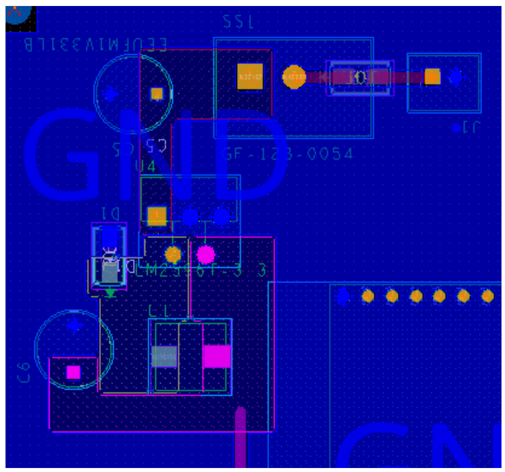


Figure 3.3.2.g: Layout of power circuit on the first prototype PCB.

3.3.3 Audio and Electronics System Validation and Testing

To test if the device was capable of outputting audio at 80 dB SPL at 1m, the first test was done to confirm the SPL requirement in conjunction with measuring the current draw of the MP3 Player while playing different treatments. This was done using the experimental setup shown in **Figure 3.3.3.a**, where the MP3 player was supplied with 3.3V from a DC power supply, measuring that supply current using an ammeter.

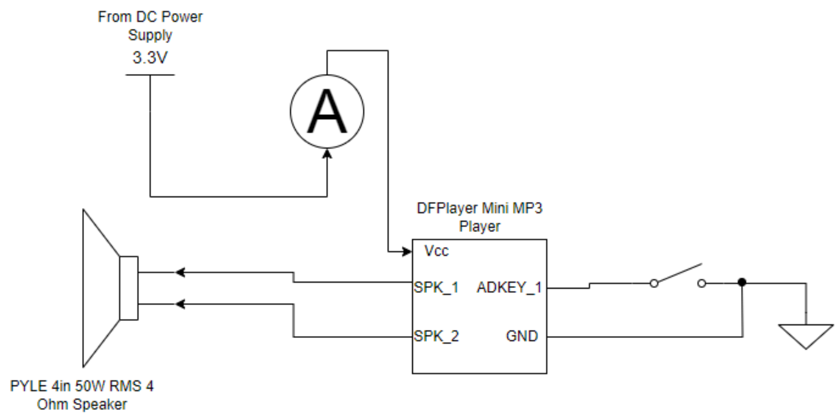


Figure 3.3.3.a: Schematic of circuit used to test the MP3 current consumption. Grounding the ADKEY_1 pin of the MP3 player plays the first track on the micro SD card on the MP3 Player.

The treatment was played using the AD button control mode, where grounding the ADKEY_1 pin plays the first track on the micro SD card. Using the AD button control to output audio plays the track at max volume, used to confirm whether the audio setup could reach 80 dB SPL at 1m. **Table 3.3.3.a**

Control Bird		Human (Swahili)		Cattle	
Time (s)	Current (mA)	Time (s)	Current (mA)	Time (s)	Current (mA)
5	60	5	220	5	110
10	50	10	240	10	90
15	70	15	100	15	90
20	70	20	130	20	90
25	60	25	320	25	90
30	60	30	180	30	100
35	60	35	230	35	120
40	180	40	200	40	100
45	60	45	190	45	110
50	60	50	90	50	130
55	70	55	380	55	110
60	60	60	230	60	90
Average Current = 71.67mA		Average Current = 209.17mA		Average Current = 102.5mA	
Max Current = 180mA		Max Current = 380mA		Max Current = 130mA	
Min Current = 50mA		Min Current = 90mA		Min Current = 90mA	
Max SPL= 90 dB SPL		Max SPL = 90.5dB SPL		Max SPL = 85 dB SPL	

Table 3.3.3.a: Results of current consumption and SPL test. The SPL was measured using a SPL meter 1m away from the speaker.

The results from **Table 3.3.3.a** confirm that the audio system is capable of reaching 80 dB SPL at 1m and the max current consumption from the test was used by the Power Engineer to inform him at what rate to discharge the battery. The experiment also confirmed the datasheet number of the standby current being 20mA, which was also used by the Power Engineer in the Power Budget document to inform the low-state current draw. This test was used to inform the team what treatment to use to test the worst-case, as the human treatment average current consumption was over double the cattle and control treatments. The human treatment is much higher due to the audio signals having a much higher peak-to-average ratio, even though the peaks are similar.

After the first prototype PCB was milled and assembled by Aiden and Liam, a test to measure the power supply signal integrity and audio quality of the device on the PCB was done. This experiment, shown in **Figure 3.3.3.b** below, involved measuring the power supply signal on an oscilloscope using a test point on the power rail.

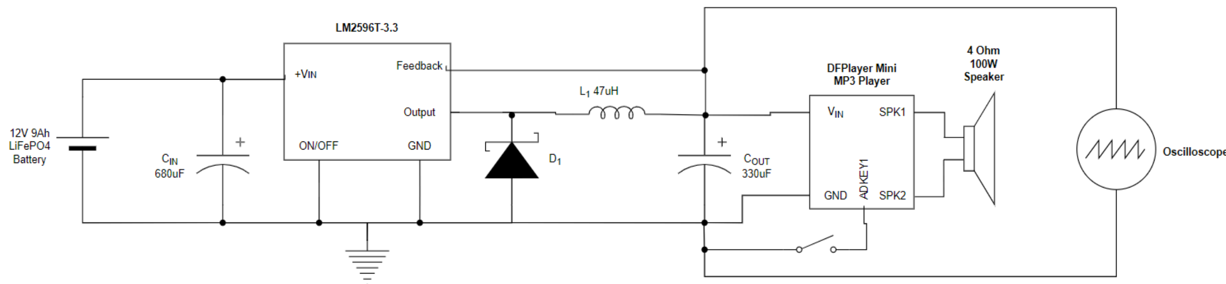


Figure 3.3.3.b: Schematic of power circuit test setup used in the power circuit experiments. Oscilloscope probes attached to the board ground and test point on 3.3V rail**.**

The test was conducted with no output audio playing from the device, which is when the power signal should be most stable, and the measured results showed a very noisy power supply signal shown in **Figure 3.3.3.c**.

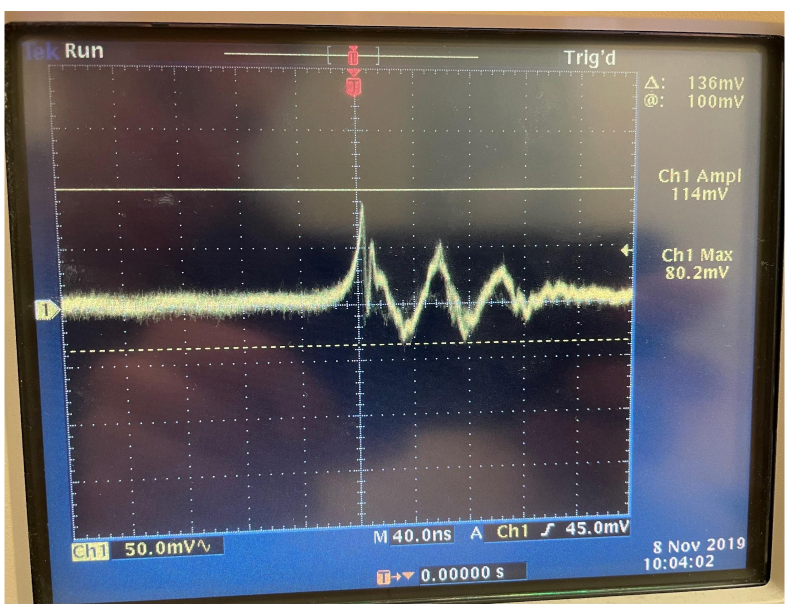


Figure 3.3.3.c: Oscilloscope measurement of audio with no audio playing. Max voltage of 114 mV above 3.3V supply.

The measurement shows a signal with a max amplitude of 114mV above the 3.3V supply voltage line, far too noisy for quality audio output. The audio qualities were then tested by playing the three different treatments at max volume. The control and cattle treatments were acceptable audio quality for the client, but the human treatment played a buzzing noise instead of a man speaking in Swahili. After supplying the power rail 3.3V using a DC power supply instead of the LM2596 power circuit on the PCB and playing the same audio track that was buzzing before, Liam and Aiden were able to determine the audio quality issue was due to the power circuit.

The problems regarding power rail signal integrity were likely due to the connections of the LM2596 switching regulator to the PCB, as the connections were made by soldering wires to the IC and soldering those wires to the board. This was done because the wrong footprint was used for the LM2596 that was ordered when designing the layout, causing the leads of the IC to not fit on the board. A new board was milled with a corrected footprint and filtering capacitors near the power pins of the microcontroller and MP3 player. After closer datasheet inspections, it was determined that the microcontroller and MP3 player needed larger bypass capacitors, another revision that will improve the audio quality of the device. Along with the already 100nF bypass capacitor used to filter out high-frequency noise, a 100µF capacitor was added on the power supply rail close to the input voltage pin to help filter lower frequency noise.

Once the second prototype board was milled and assembled by Liam and Aiden, the same experiment was run on the new board using the experimental setup shown in **Figure 3.3.3.b**. The power system was tested again first with no audio output, and the results of this experiment showed a much cleaner power supply voltage shown in **Figure 3.3.3.d**.

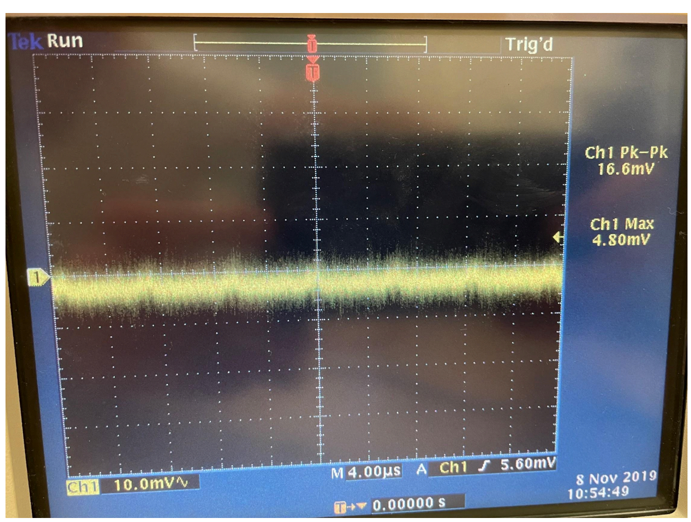


Figure 3.3.3.d: Oscilloscope measurement taken at test point on power rail with no audio treatment playing. Peak-to-Peak voltage of 16.6mV.

The measurement of the 3.3V power supply rail recorded a peak-to-peak voltage of 16.6mV, much lower and more stable than the power supply on the first prototype board. Next, the power supply rail was measured with the highest peak-to-average audio treatment playing at max volume, which is the Human (Swahili) treatment. The results of this measurement are shown in **Figure 3.3.3.e**, show a noisier power supply signal, but still much lower than the results of the experiment on the first prototype PCB.

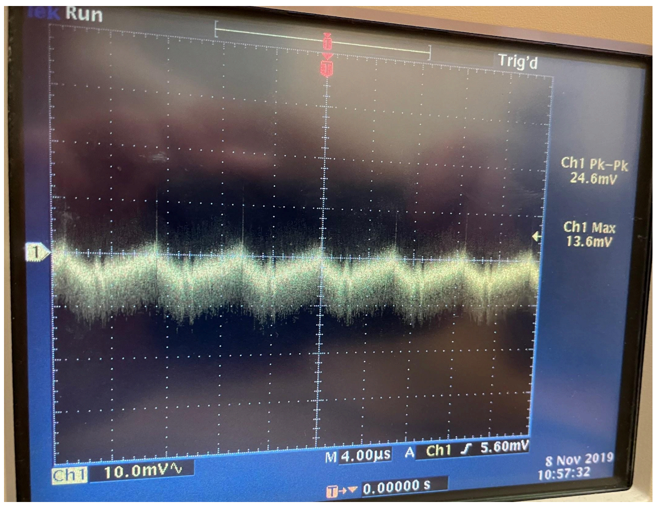


Figure 3.3.3.e: Oscilloscope measurement taken at test point on power rail with Human (Swahili) treatment playing. Peak-to-Peak voltage of 24.6mV

This test of playing the Swahili treatment at max volume had no buzzing or clipping when testing. This PCB prototype was then ready for review by the client for approval of audio quality. To get client approval on quality, multiple Human treatment tracks were played at maximum volume. After testing the different treatments, the client decided that the quality was satisfactory to be used in a playback device in Africa.

3.4 - PCB and Packaging Systems [Aiden Krauss]

3.4.1 - Subsystem Planning and Block Diagram

The PCB and Packaging Subsystems, led by Aiden Krauss, is responsible for integrating all relevant components into a consolidated design. All electronic components requiring a medium to communicate with one another will be integrated onto a printed circuit board (PCB) and the remaining components will be integrated into a casing. This subsystem has several deliverables for both the PCB and casing. The casing design requirements will be discussed later in section 3.4.6. In terms of the PCB, the following requirements were met:

1. The Microcontroller, MP3 Player, Buttons, and DC-DC Buck Converter Circuit will all be integrated on a PC Board and be able to communicate with one another seamlessly.

2. A DC-DC Buck Converter Circuit will be constructed that is more power efficient than the previous design (>73%) to conserve power and convert the battery voltage to 3.3V to power the Microcontroller and MP3.

3. The signals between the Microcontroller, MP3 Player, and Speaker will have stable signals to not allow excess noise and distort the audio playback, as evaluated by Mike Kowalski.

4. The PCB design will be possible to replicate by the average person; all parts will be possible to purchase online and soldered to the PCB with the use of a soldering iron.

To meet these requirements, several design decisions were made in collaboration with Liam, Cole, Kamran, and Mike. The design process of each PCB prototype was done in collaboration with Liam. The block diagram for the PCB is as follows:

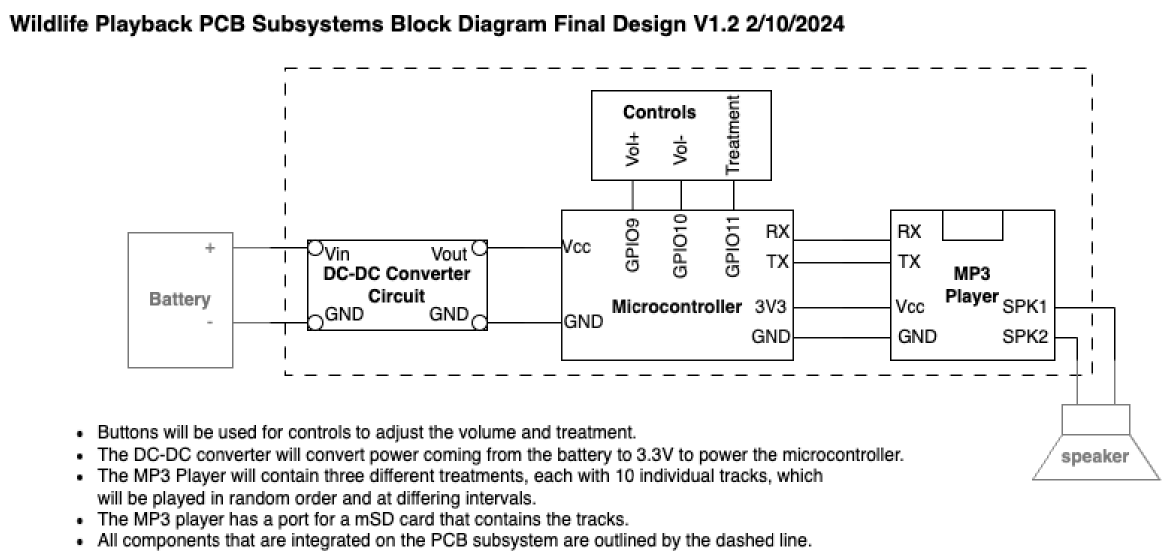


Figure 3.4.1.a: Wildlife Playback PCB Subsystem Block Diagram

The components shown inside the dotted line in **Figure 3.4.1.a** above were constructed and integrated onto the PC Board. Power is supplied to the PCB via the battery, which is delivered to the DC-DC Buck Converter Circuit (as discussed in section 3.3.2) to be stepped down to 3.3V. This then supplies said voltage to the ESP32 Microcontroller and MP3 Player. The microcontroller has three buttons connected to GPIO ports 9, 10, and 11 to allow for volume increase, volume decrease, and treatment adjustment, as programmed by Cole in the Embedded

Subsystem. The Microcontroller also communicates with the MP3 via GPIO17 and GPIO18, functioning as RX and TX communication lines. Finally, The MP3 Player is connected via headers to the speaker, which is mounted on the lid of the case.

Part choices were relatively straightforward, with the DC-DC Buck converter components selected based on the Texas Instruments datasheet discussed earlier. The ESP32 Microcontroller was selected from a previous recommendation by Cole and Mike, based on its communication capabilities, versatility, and established code library. The DFPlayer Mini MP3 Player was selected based on storage size of up to 32GB, compact size, and ease of installation within an embedded system. Basic buttons were selected due to their high adaptability in several PCB designs, and straightforward use. Extensive collaboration with the other three systems was required in building an engineering schematic. Collaboration with Liam was vital in designing the DC-DC Buck Converter Circuit, as his knowledge was vital to ensuring we had all the components necessary. Collaboration with Cole was needed in deciding what GPIO pins to use for communication between the buttons and the MP3 Player. Mike was consulted throughout the design process to ensure the design met his in-situ needs, in terms of ease of adjustability and robust construction to prevent premature degradation of the design.

3.4.2 - Engineering Schematic and Layout

The first stage of the development of the PCB design was to create an engineering schematic of the entire circuit. This was done on OrCad Capture 17.4. This schematic ensured that all components were accounted for and acted as a living document, being updated whenever necessary design changes were called for. The first prototype engineering schematic is shown below in **Figure 3.4.2.a:**

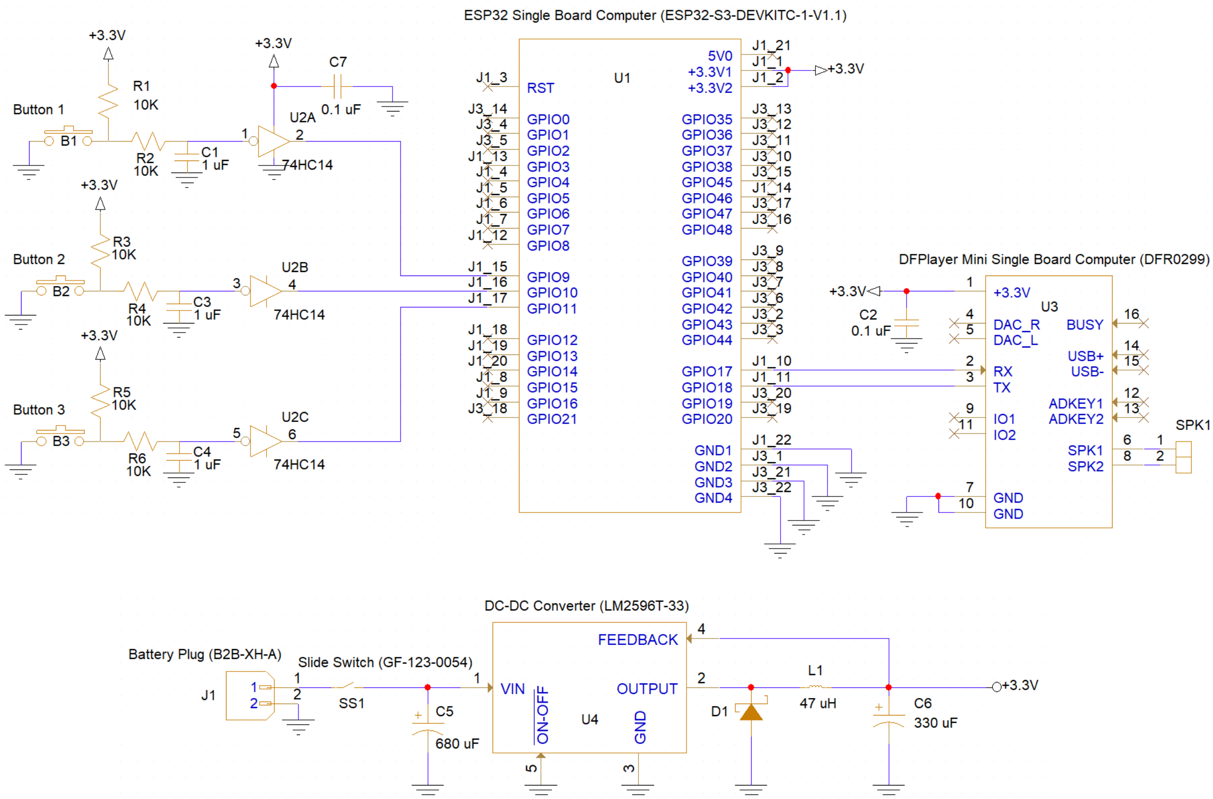


Figure 3.4.2.a: First Prototype Engineering Schematic

There are two main components to the schematic: The Audio Playback Circuit and Power Circuit. The Audio Playback Circuit contains the button controls, ESP32 Microcontroller, MP3 Player, and Speaker connection. The Power circuit contains the Battery connection and DC-DC Buck Converter Circuit.

The most notable component of the design is the choice of physical hardware button debouncing, as opposed to the alternative software method. The concept of debouncing a switch comes from the fact that most switch contacts are not stable when first connected, and often fluctuate between high and low for a certain time after the button is pressed before the signal stays at a steady high. This fluctuation may be registered by the input connected to them and cause tens of undesired switch triggers [A].

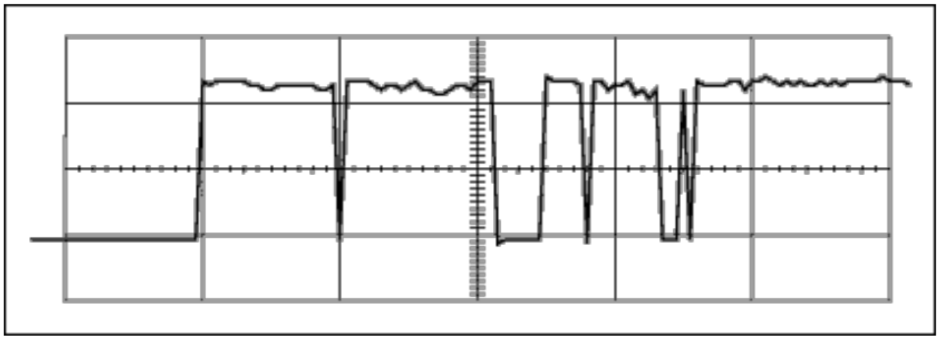


Figure 3.4.2.b: Undesired switch triggers requiring debounce before a steady signal. Time progresses from left to right, with the button being pressed only at the first rise of the signal.

This circuit design used in this first prototype was derived from the Texas Instruments: Debounce a Switch datasheet [B]. A basic circuit was recommended in the datasheet, along with a table of resistor value choices based on the desired debounce time constant. The typical time constant used is around 10ms, so component values were selected to account for this [B]. An additional resistor is shown in our debouncing circuits representing the built-in pull-up resistor from the microcontroller.

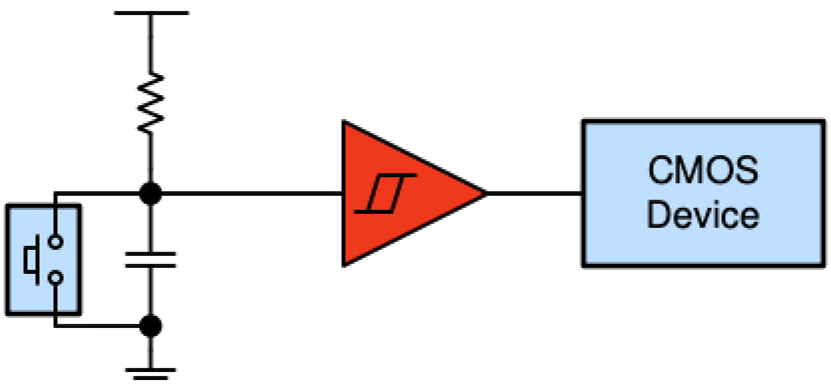


Figure 3.4.2.c: Hardware-based button debouncing circuit, as shown in TI Datasheet [B].

Time Constants for Common Resistor and Capacitor Values

Capacitor (μF)	Time Constant (ms)(1)		
	R = 10 kΩ	R = 100 kΩ	R = 1 MΩ
1	**10**	100	1000
0.1	1	**10**	100
0.01	0.1	1	**10**

Table 3.4.2.a: Capacitor/resistor choices based on time constant values shown in TI datasheet [B].

In terms of the Power Circuit, the Buck Converter Circuit was derived via the previously mentioned LM2596 Simple Switcher Texas Instruments Datasheet. This circuit required extensive research and discussion with Professor Petersen to understand the function of each component in the circuit. The design continued across all prototype designs and proved to be adaptable to the changing conditions of our circuit designs and tests.

After the development of the schematic was complete, the next stage of laying out the circuit began. The first prototype layout phase took much longer than anticipated, due to the amount of issues encountered. Liam and Aiden collaborated in designing the Schematic and Layout. Both had previous experience with PCB design in ECE 173, but was based on previously established designs, and lacked experience in creating new designs from scratch. The top and bottom sides of the final layout of the first prototype are shown in **Figure 3.4.2.c** and **Figure 3.4.2.d** below:

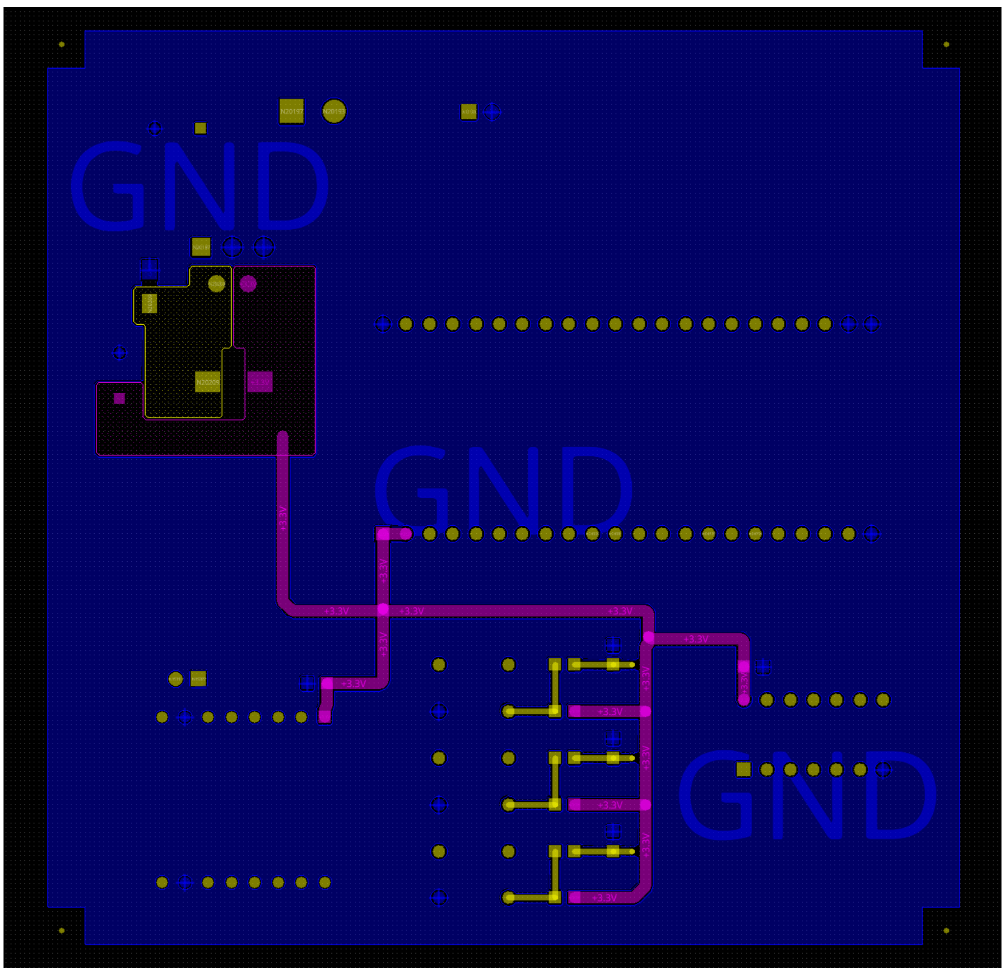


Figure 3.4.2.c: First Prototype Layout Top Layer

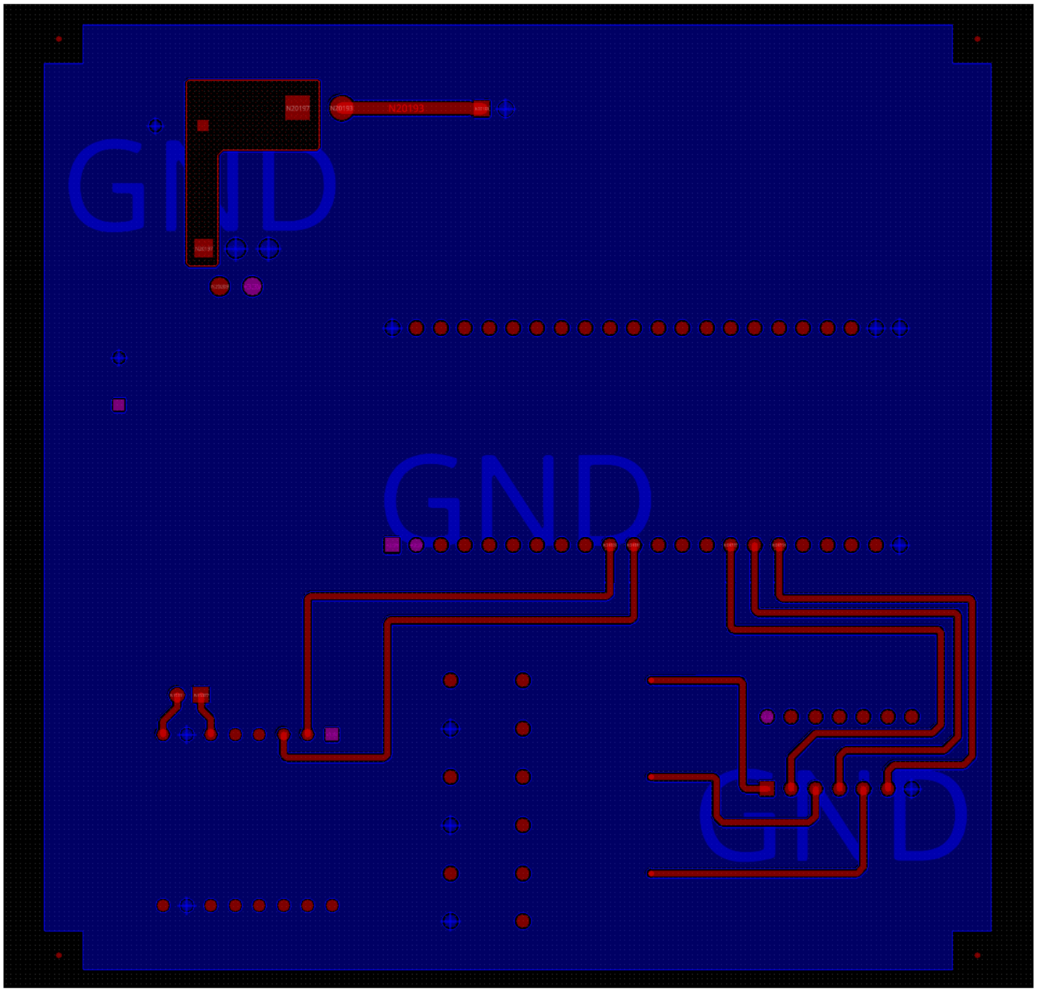


Figure 3.4.2.d: First Prototype Layout Bottom Layer

Both layers contain a large ground plane, as shown by the blue shape covering most of the design. The Buck Converter Circuit is shown in the top left corner of the design, integrated across both layers with the red, yellow, and pink copper areas and parts connected to them. The Microcontroller is represented by the two horizontal lines of 22-pin holes, which are connected to the MP3 Player on the bottom left, and the button debouncing circuit on the bottom right. The pink line across the top layer of the board represents the power rail from the buck converter, supplying power to the MP3 Player, microcontroller, and inverter used in the button debouncing circuit.

Several issues were encountered during the design process of the board layout. The most difficult to work through was creating a sufficient design of the buck converter circuit. Numerous design considerations were made, based on application notes recommended by Professor Petersen. The directional placement regarding all other components in the circuit was vital to ensuring the converter functioned as efficiently as possible. For example, the inductor, output capacitor, and output diode needed to be as close to each other as possible to reduce EMI from power traces, lead inductance, and resistance. The grounds of the IC, input capacitor, output capacitor, and output diode also needed to be as close together as possible. All high-current power traces needed to be short, direct, and thick, and were eventually replaced with the copper areas shown. The switching current loops were also required to curl (flow) in the same direction on the board [C]. All of these design considerations recommended by the application notes were accounted for and appeared in the first prototype design. Per recommendation by Professor Petersen, parts were rearranged to ensure components requiring stable signals, namely the microcontroller and MP3, to ensure their ground returns were as close as possible to the ground of the battery. Locating correct footprints proved to be more difficult than anticipated, and several modifications to existing footprints were made to account for design choices.

3.4.3 - First Prototype Construction

The construction of the first prototype board was not without its extensive list of issues encountered. The first board was made on the M60 Mill with Professor Petersen on April 17th. Construction began the next day, and with each of us relearning how to solder, took an extensive amount of time. As per Professor Petersen's instructions, the buck converter circuit was constructed first and tested to ensure the correct power would be supplied to all other components as to not harm them upon installation. Our first prototype board with the power circuit and microcontroller mounted is shown in **Figure 3.4.3.a:**

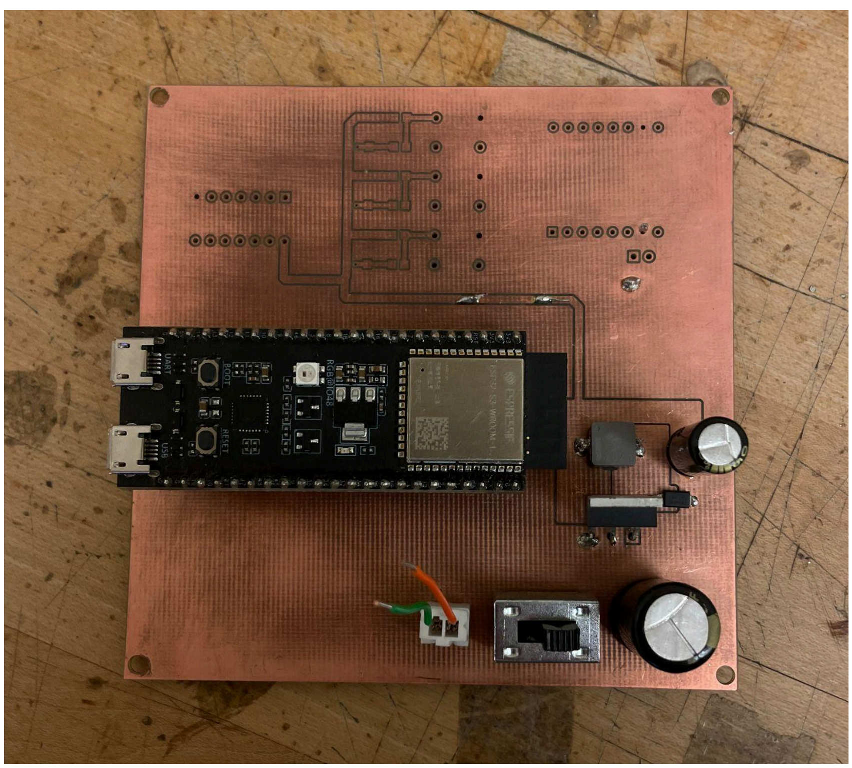


Figure 3.4.3.a: First Prototype Board with Buck Converter Circuit and Microcontroller Installed.

Extensive testing was done to work out bugs in our power circuit. For an extensive period of time, our output voltage was half of our input voltage, as opposed to the required 3.3V. After consultation with Professor Petersen and endless searching, we found that the footprint for our LM2596 IC was mirrored incorrectly, and the output and feedback pins were positioned in the wrong direction. We removed the IC and reconnected it correctly with jumpers, which finally provided us with the correct output voltage. However, our efforts to prevent harm to our microcontroller and MP3 player proved to be futile, as our connections were unstable and output voltage fluctuated. With careful handling and resoldering, we were able to establish what we thought was a stable power supply to the rest of the board. However, when we mounted the MP3 and speaker, we encountered a new issue. When playing certain high-current audio tracks, a buzzing noise would be heard instead of actual audio, which would eventually fade and turn off. Puzzled by this issue, we performed more testing on the circuit and eventually discovered that the wires we connected the LM2596 with were introducing unwanted noise into our power supply, making it impossible for the MP3 to play tracks. Among other issues encountered, we decided it was best to scrap this prototype and create a new design.

3.4.4 - Revisions/Second Prototype

Given the problems we encountered, we added several changes to our second prototype.

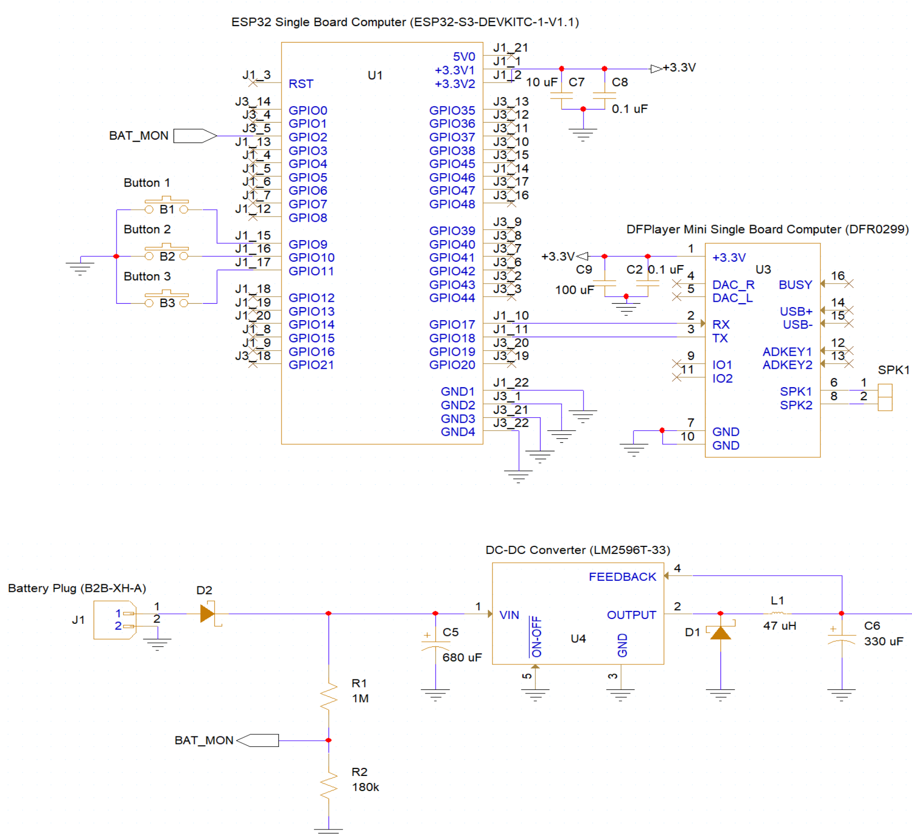


Figure 3.4.4.a: Second Prototype Engineering Schematic

There are several notable improvements made to our second schematic. Our physical hardware debouncing circuit was removed and replaced with software debouncing to cut down on the number of parts, costs, and construction time. Filtering capacitors were added to both the ESP32 and MP3 to improve the quality of the power supplies, in hopes of solving the audio buzzing issues. A battery monitoring circuit via a voltage divider was added to add the ability to track battery levels across time for testing purposes. A protection diode was also added directly after the battery plug to protect the circuit in the event of an incorrect battery connection. With a significantly different schematic and circuit, notable changes were also made to the layout shown below:

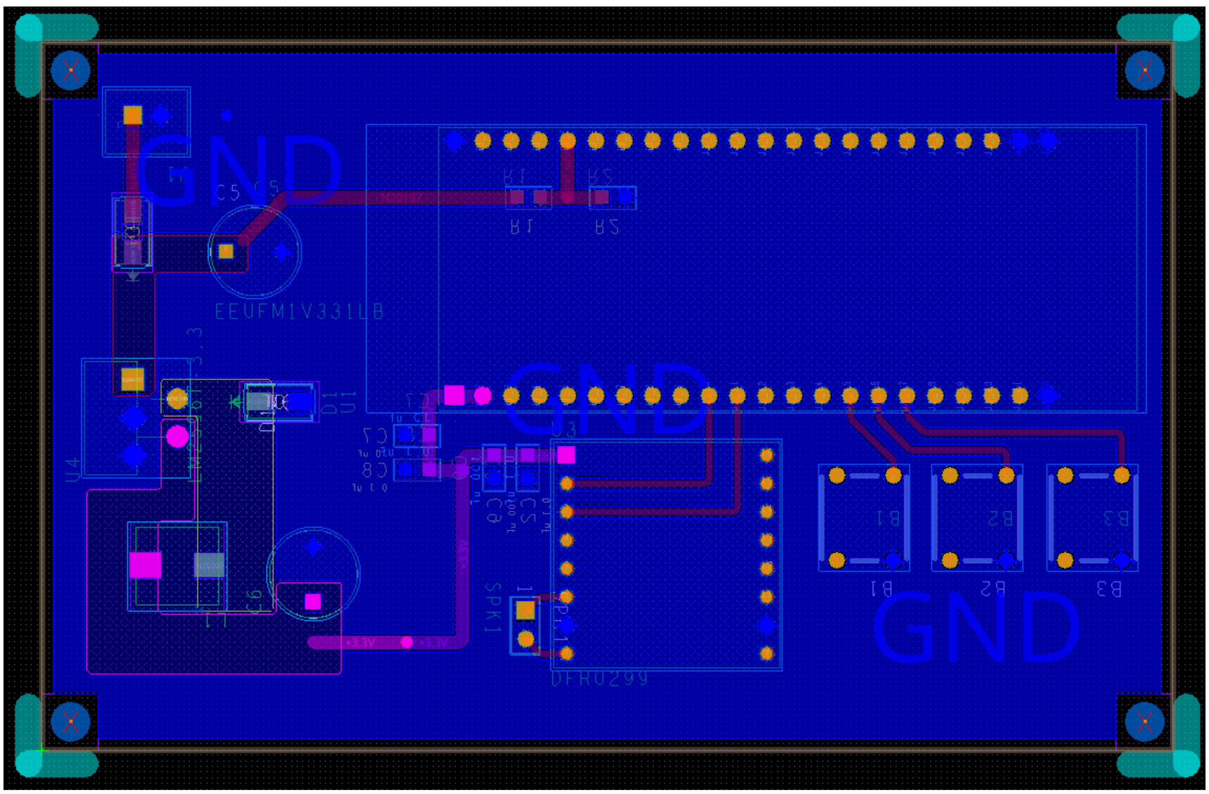


Figure 3.4.4.b: Second Prototype Board Layout

The figure above shows both the top and bottom layers, with traces and components on the bottom layer shown in red, and traces and components on the top layer shown in yellow and pink. Both sides continued to have a ground plane across the entire layer to ensure proper ground returns. The LM2596 footprint was correctly mirrored on the second prototype and could be mounted correctly without the use of wires. The power circuit was flipped across the horizontal axis, with the same design considerations discussed previously, but in a more compact design. With the button debouncing circuit removed, the board was able to be condensed down into a 2.5” by 4” board, as opposed to the previous 4” by 4” board. The battery monitoring circuit can be seen as the red trace traveling from the power circuit on the left under the microcontroller, and connected to GPIO2.

During the board construction phase, further issues with the power circuit were encountered, with the feedback trace having a loose connection, causing the output voltage to mirror the input voltage. However, when testing and pressing the pin with a multimeter and causing the loose connection to reconnect, we were unable to locate the issue using continuity tests or resistance tests. Eventual resoldering of the LM2596 solved the issue, and a correct and stable 3.3V power supply was established for the rest of the board. The addition of filtering capacitors and direct connection of the LM2596 to the board solved the audio buzzing issue as well. However, this second board was milled by someone other than Professor Petersen with less experience, and milling and drilling tolerances were incorrectly calibrated. This caused “flashing”, where the drilling or milling bit would cause the traces to fray at the edges, and resulted in traces ripping off the board, ruining some of our connections. Though our board functioned properly as per previously discussed design deliverables, it proved to be unreliable, with multiple traces needing multiple resolders or bridges with wires. Our finished second prototype board is shown below:

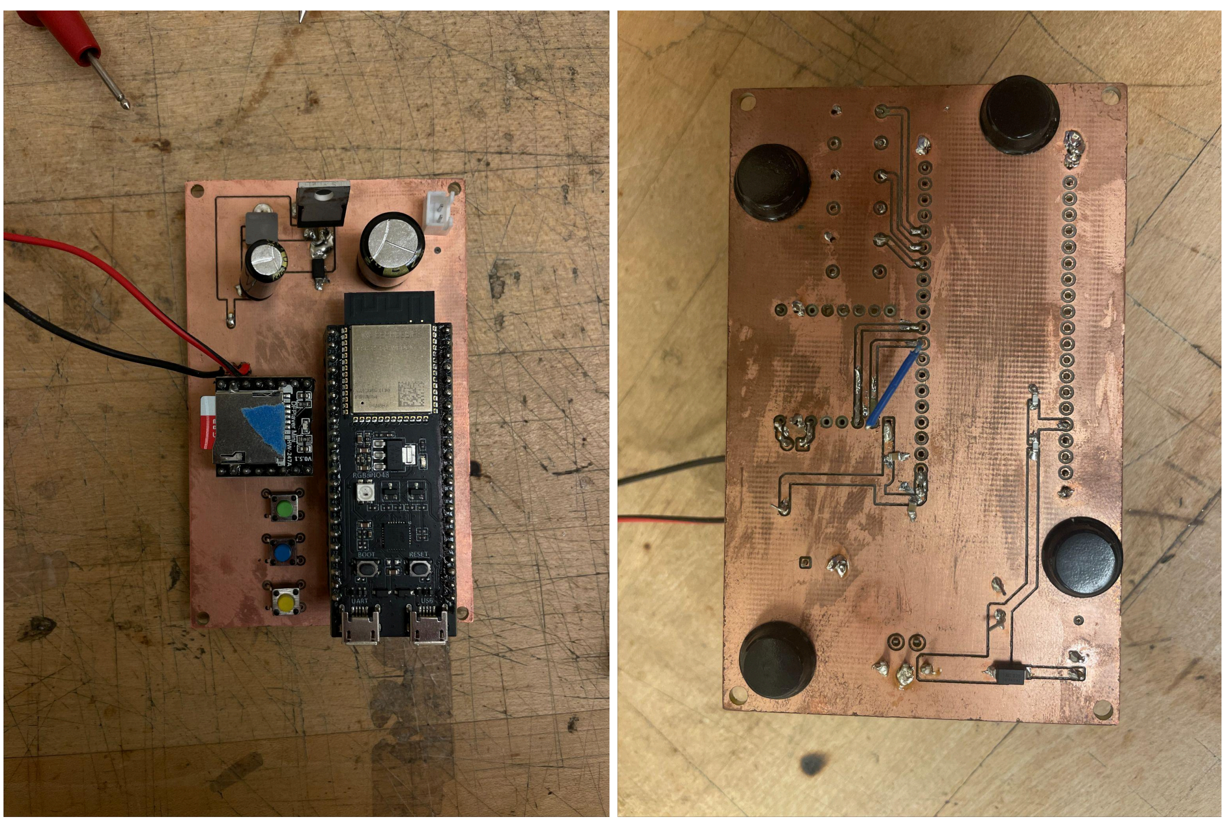


Figure 3.4.4.c: Finished Second Prototype Board, Functions Successfully.

3.4.5 - Third Prototype/Current Design

After creating a successfully functioning prototype, we moved on to considering Mike's need for mass production of these boards for his fifty devices. After extensive research and discussion of the best method to assemble sixty of these boards (fifty for the devices and ten backups), we decided it was best to have the boards milled by JLCPCB, and constructed by one of Kamran's Father's previous colleagues. Our Engineering schematic remained the same as the previous design, but we realized we may need layout changes in preparation for production. We consulted Kamran's Father, who has worked in the electrical engineering field for over 25 years, about design revisions. Our third prototype layout is shown below:

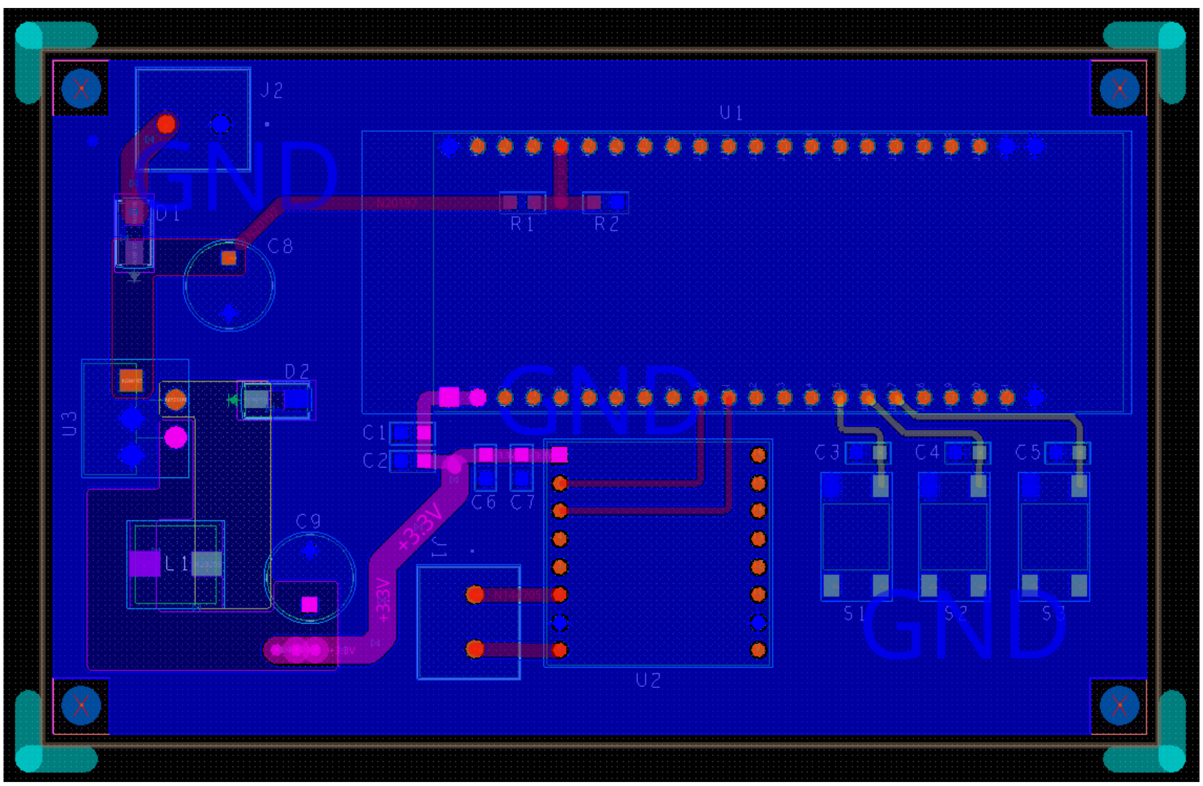


Figure 3.4.5.a: Third Prototype Board Layout

The most notable changes include widening the power traces in reference to Standard IPC-2221, which states that typical one and two-ounce copper thickness traces should account for ten mils per one amp of current flowing through them [D]. Given our maximum current rating of three amps, we ensured that all high-current power traces were at least thirty mils thick. We were also instructed to further widen the traces from the battery to the power circuit, and the power circuit to the MP3 and ESP32 power rail junction, up to 100 mils. The choice of buttons was also changed from through hole to surface mount to allow for easier board construction. The battery and speaker connections were replaced with wire-to-board connectors, so thicker wires could be used and clamped down, allowing for more reliable connections.

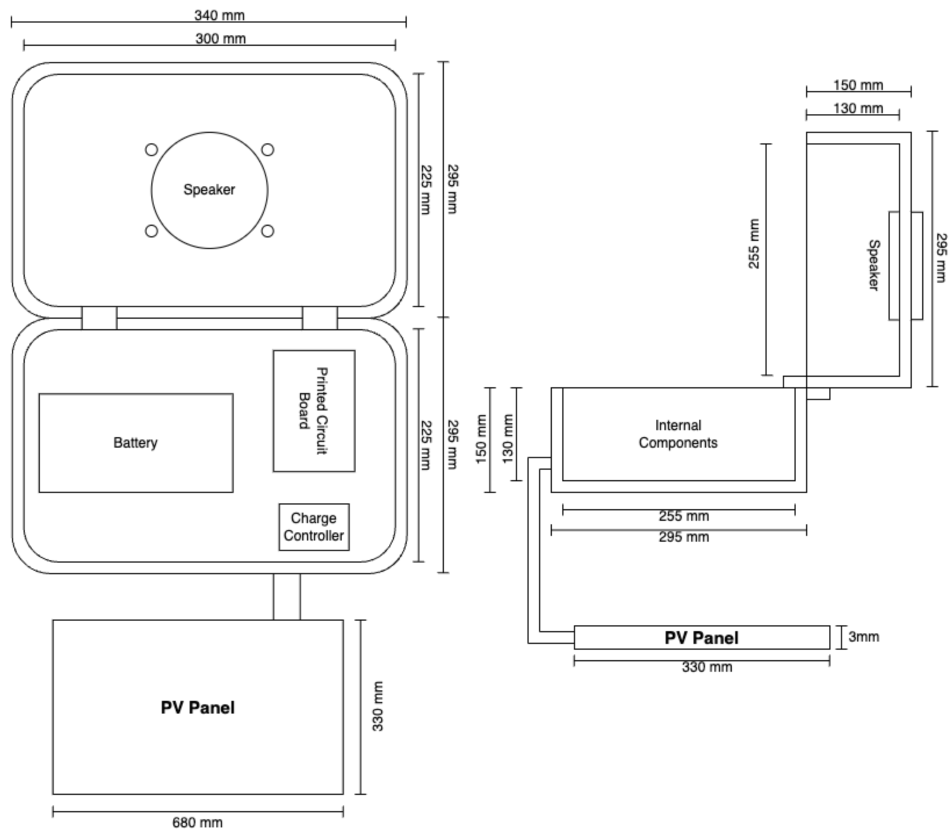
However, with the time required to change the layout, order all parts necessary, and order and receive the PCBs, our original plan to have the colleague assemble our boards fell through. She was set to leave for a month on June 12th, which is only eight days before Mike is required to depart for Kenya. We received the boards and parts on June 11th, so we were unable to bring them to her in time. As of now, our current plan is to assemble the boards ourselves, as we have four of us on our team, and each of us has time available before June 20th to assemble 15 boards each.

3.4.6 - Packaging Design

Finally, as for the encasing of our design, there were several design requirements for the case, as shown below.

1. The casing must be a certified NEMA Type 4 Enclosure, to prevent ingress of water and solid foreign objects, and protect wildlife from harmful components inside, as discussed earlier in section 2.3: Validation of Requirements.
2. The casing must be a light tan color to blend in with the surrounding environment and reflect sunlight energy to keep internal temperatures lower than the previous black casing.
3. The casing must have a smooth surface on the top side large enough to mount the speaker, which is a 5.25” diameter circle requiring a 4” diameter hole.
4. The casing must be able to have a python lock mounted to it, with at least two ⅜” holes drilled into the bezels to thread the python lock through to securely mount the device on a tree.
5. The casing must be large enough to fit the battery, PCB, and charge controller inside while keeping the size down to allow for convenient transportation from the United States to Kenya.

These design deliverables changed slightly throughout the process of designing the rest of our devices. The decrease in board size influenced the size of the case needed, so a final design for a case was not chosen until late in the project. Finding a tan case with a smooth lid turned out to be more difficult than expected, so a design decision was made to purchase a black case with a smooth lid and repaint it a tan color. Many cases had bezels or ridges that rendered it impossible to mount the speaker without extensive modification, which would use up valuable time in the end stages of our project. Eventually, after a discussion with Mike, an economically viable option was chosen. The Mechanical Drawing of the packaging design is shown below:



Version Control			Wildlife Playback Casing V1.0 4/26/2024
Version	Date	Description of Changes	
0.1	2/4/2024	Initial Draft of Casing Subsystem Block	
0.2	2/15/2024	Revision to Mechanical Drawing	
1.0	4/26/2024	Revision to Final Component Selection	

Figure 3.4.6.a: Mechanical Drawing of Packaging System

The Mechanical Drawing proved to be rough guidelines for the placement of each component within the casing itself but was difficult to follow when it came to building the actual device. Many changes were made to the orientation of the components before the final orientation, as shown above. As for the case chosen and the actual device, the figures below show the top side, bottom side, and internals of the device.

Figure 3.4.6.b: Final Casing Choice, Top side with Speaker Mounted

Figure 3.4.6.c: Final Casing Choice, Bottom Side

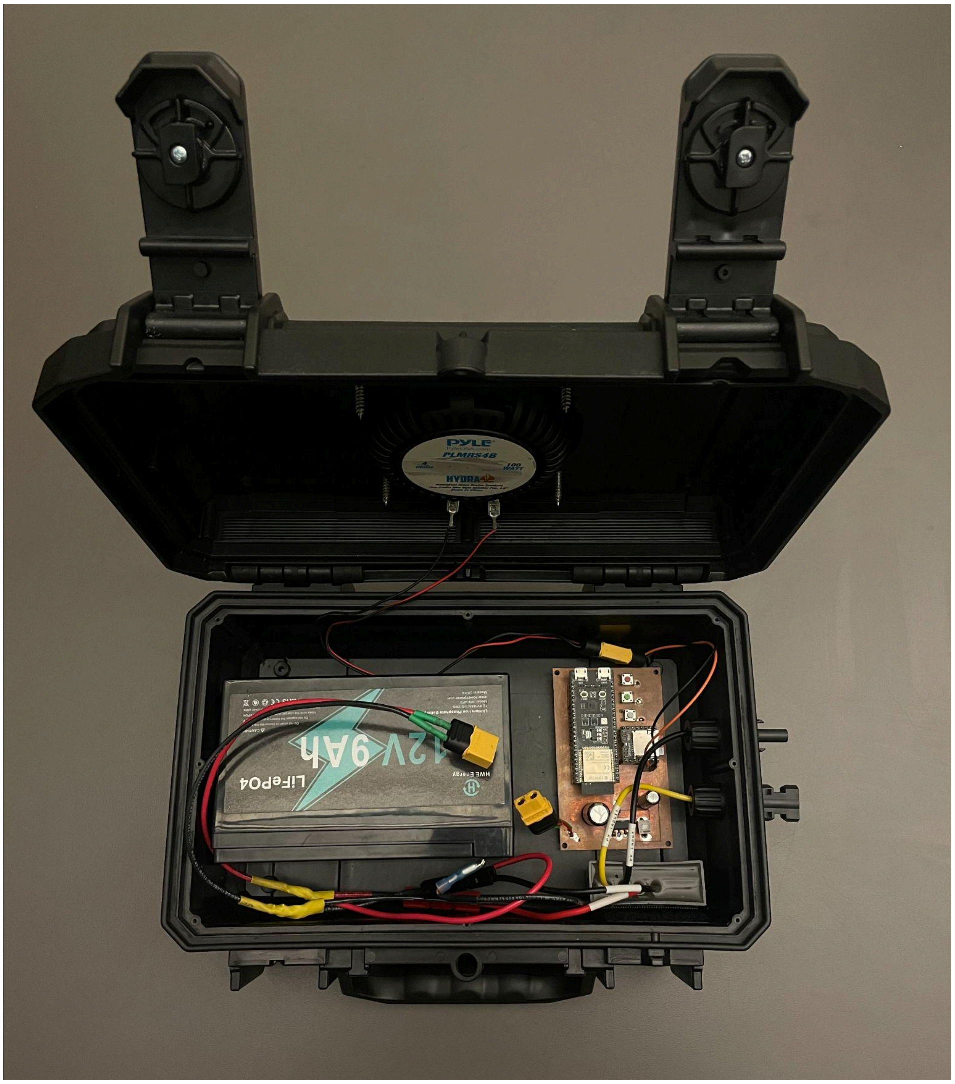


Figure 3.4.6.d: Final Casing Choice, Inside View

It was elected to mount our solar panel plugs on the right side of the box, which would later face up when mounted to a tree, allowing the solar panel to be placed above the device. **Figure 3.4.6.d** above shows the final orientation of the interior components, with the battery velcroed to the bottom to ensure it does not fall on top of other components, the PCB velcroed on the bottom, and the charge controller mounted to the side. The final design with the speaker and solar panel plugs is still water resistant, as proved by tests conducted by spraying the casing again with a twenty-second shower hose and checking the interior for any water ingress. Our current design for the casing choice and orientation remains unchanged and will be used in the field for Mike's experimental devices.

4 Lessons Learned [Cole Quinlan]

Throughout the development of the group's device project, the team gained invaluable insights and lessons that significantly influenced the design and implementation process. From the initial concept to the final prototype, each phase presented unique challenges and learning opportunities. The discovery of the importance of troubleshooting, application notes, and good engineering practices ensured the functionality and reliability of this product. The integration of components in subsystems and the integration of subsystems into the final prototype required a holistic approach to design, which could only be improved by the continuation of good practices. Additionally, effective communication and collaboration within the team proved essential in overcoming technical obstacles and refining our design. These lessons not only enhanced the group's technical skills but also underscored the importance of adaptability and continuous improvement in this engineering project.

Troubleshooting, a large part of the group's success, played a pivotal role in identifying and resolving issues in every subsystem. For example, when the device's battery monitoring system failed to provide accurate readings, systematic troubleshooting allowed the engineers to identify a faulty ADC pin configuration as the root cause. By addressing this early, the team prevented future problems that could have impeded the device's functionality and performance. Similarly, during the initial UART Connectivity Test, the group encountered connectivity issues between the microcontroller and the MP3 module. Systematic troubleshooting revealed a faulty connection on the Rx pin of the MP3, which was then resolved by re-soldering. This process ensured that potential flaws were addressed early in the design phase and did not cause significant issues later in the development cycle. Moreover, a solid troubleshooting routine provided valuable insights into the design process, guiding improvements and optimizations that elevated the overall quality of the device. Without a solid understanding of troubleshooting, completing a high-performing, dependable, and user-friendly final product would have taken much longer. Troubleshooting becomes significantly more effective when reinforced with application notes that help compare values against differing results.

Complementing the troubleshooting process, application notes were equally crucial in the development phase of the device. For instance, when integrating a new sensor, the manufacturer's application notes provided detailed guidelines on optimal use, including recommended wiring diagrams and example code. These documents offered practical information on typical applications, best practices, and potential pitfalls. The value of application notes was evident from reference designs, schematic diagrams, and example codes that significantly accelerated the development process. For this group, they assisted in understanding the intricacies of new technologies, ensuring components were used correctly and efficiently. For example, the ESP32-S3 microcontroller's application notes provided essential details on configuring its ADC pins and setting up UART communication, which were crucial for accurate voltage monitoring and reliable data transmission. By leveraging the knowledge contained in application notes, the team avoided common mistakes and optimized performance with each part. To optimize both troubleshooting and application note lessons, good engineering practices helped to refine every decision the group made.

Adhering to rigorous engineering standards was fundamental to the successful design and development of the device. These standards encompassed a range of best practices and protocols that ensured the safety, reliability, and efficiency of the final product. For instance, the group maintained thorough documentation of all design and development processes, which facilitated clear communication and future troubleshooting or modifications. Employing modular design principles enhanced maintainability and scalability, allowing for discrete testing and easier updates to individual components, as demonstrated when the team upgraded the power management subsystem without affecting other parts of the device. Rigorous testing and validation procedures, such as unit testing, integration testing, and automated testing frameworks, were critical for identifying and resolving issues early in the development cycle. The utilization of version control systems like Git ensured efficient tracking and management of code changes, preventing conflicts and data loss. Adherence to industry standards and regulatory requirements, including ISO 9001 for quality management systems and IEEE standards for electrical and electronics engineering, ensured compliance and promoted best practices throughout the project lifecycle. Implementing systematic design methodologies, such as the V-model for verification and validation of Agile methodologies for iterative development, structured the development process and enhanced team collaboration. Moreover, conducting thorough design reviews and peer evaluations at each stage of development helped identify potential issues and ensured adherence to design specifications. By upholding these advanced engineering practices, the team's engineers produced high-quality, durable, and user-friendly results in the form of the wildlife playback device. This rigorous approach not only enhances the performance and reliability of the device but also reinforces the credibility and reputation of the engineering team and their product, ensuring sustained success in the real world.

In conclusion, the project's success was a result of integrating lessons learned in troubleshooting, application note utilization, and adherence to good engineering practices. Each of these aspects played a crucial role in navigating the complexities of the wildlife playback device, from the initial concept to the final prototype. The structured approach to identifying and resolving issues, leveraging detailed component guidance, and adhering to industry standards ensured the development of a reliable and efficient device. These practices not only put the final products on a professional level of engineering but also prepare the engineers going through this course for the real world.

5 Conclusion and Next Steps [Kamran Pakravan]

5.1 Conclusion

By completing this project, the team has helped a graduate student, Mike Kowalski, be able to gather the research needed to better understand human impact on the environment. After showing the device and its functionality to Mike, he was very impressed and pleased with the design. When compared to his initial prototype, it is clear that our device is a significant improvement. Increased durability, battery life, the ability to change tracks and volume, as well as being able to change the duty cycle, are all requirements set by Mike that we were able to successfully deliver. Because almost all the parts of the project are open source (minus the PCB), this project is also repeatable. This means that the device can be made and used by other environmental scientists hoping to observe the effects of human disturbance on their local environment. The data and information Mike collects using this project have the potential to influence how the environmental science community views multi-use landscapes as a potential solution to the encroachment of protected areas.

5.2 Manufacturing Experimental Devices

The next steps for this project is to help Mike mass produce these devices to be able to complete his experiments in Kenya. Mike aims to produce 50 audio playback devices in the next week. Steps are being taken to help him manufacture all these devices. As of now, all the components required to manufacture the devices have arrived. Mike has built all of the wiring harnesses required for the project and has begun to drill out the boxes and build the physical devices. The PCB boards and components have all arrived, and we aim to have a boardhouse solder our components onto the board. Solar panels have also been ordered and are shipping to Kenya. One challenge that we faced was obtaining all of the required components on time. Components in large quantities took longer to ship than expected, which led to some setbacks. Another challenge that Mike is trying to tackle is dealing with the battery providers in Kenya. The providers in Kenya are not as responsive and reputable when compared to suppliers in the United States. However, Mike has experience with these suppliers and has friends in Kenya who ensure that they will be able to obtain the batteries on time. We are working diligently to help ensure Mike can complete his project in Kenya.

5.3 Future Revisions

If a future team of engineers were to tackle this project to make improvements, or if this team had more time, making the device able to communicate with phones through the RF or Bluetooth functions would be extremely beneficial, as expressed by the client. A remote monitoring system that would allow users to view the battery life, the current track playing, and the logs of the device’s charge and discharge data would make life for the user even easier.

Implementation of remote adjustment for the device would save the user multiple hours, as they would no longer have to travel to each device individually to change the duty cycle, volume, or treatment. A custom-designed case would have also been ideal for the project given more time. 3D-printed internal separators would improve the organization and cleanliness of the design. They would also ensure that components were fit even more snugly, preventing any accidental damage during transportation. Given even more time, a playback speaker device with a built-in motion sensor game camera would be able to converge multiple devices used in the experiment into one device.

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